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MILITARY SPECIFICATION

SUBMARINE INTERCOMMUNICATION SYSTEM

AN/WIC- (V)

1. SCOPE

1.1 This specification covers an amplifier intercommunication system for submarines to provide for the transmission of voice, call signals and electronically generated alarm signals to the various compartments and operating positions in a submarine. The AN/WIC intercommunication system consists of microphones, handsets, headsets, loudspeakers, audio amplifiers, signal generators, power supplies, control equipment, wiring junction boxes and test facilities.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-1 Electron Tubes and Crystal Rectifiers.
MIL-R-22 - Resistors, Variable (Wirewound, Power Type).
MIL-C-25 - Capacitors, Fixed, Paper-Dielectric.
MIL-T-27 - Transformers and Inductors, Audio, Power and Pulse.
MIL-R-94 - Resistors, Variable, Composition, General Specification for.
MIL-S-901 - Shockproof Equipment, Class HI (High Impact) Shipboard Application, Tests for.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for.
MIL-R-2033 - Relays for Naval Shipboard Electrical Service.
MIL-S-3786 - Switches, Rotary (Circuit Selector, Low Current Capacity).
MIL-S-3950 - Switches, Toggle.
MIL-C-3965 - Capacitors, Fixed, Electrolytic (Tantalum), General Specification for.
MIL-B-5423 - Boots, Dust and Water Seal (for Toggle and Push Button Switches and Rotary Actuated Parts.
MIL-R-5757 - Relays (Electrical (Excluding Thermal), for Electronic and Communication Type Equipment, General Specification for.
MIL-S-6743 - Switches, Push Button and Limit.
MIL-S-8805 - Switches, Sensitive and Push, Snap Action, Actuators and Enclosures.
MIL-M-10304 - Meters, Electrical Indicating, Panel Type Ruggedized.
MIL-C-10544 - Connectors, Plug and Receptacle Electrical, Audio, Waterproof, Ten Contact, Polarized.
MIL-L-12606 - Loudspeaker, Permanent Magnet (Unencased, 4-Inch Diameter Cone, 2-Watt, Gunblast and Immersion-Resistant Type).
MIL-F-15160 - Fuses, Instrument, Power and Telephone.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-S-19500 - Semiconductor Devices, General Specification for.
MIL-A-20222 - Announcing Equipment, Shipboard.
MIL-L-25547 - Loudspeaker, Magnetic LS-211-AIC-13.
MIL-H-26312 - Headset-Microphone H-157-AIC.

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STANDARDS

MILITARY

MIL-STD-102 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-701 - Preferred and Guidance List of Transistors.
MS15544 - Lamps, Incandescent, T-1-3/4, Single Contact Midget Flanged.
MS25237 - Lamps, Incandescent, Midget Inverted Flange Base.

DRAWINGS

BUREAU OF SHIPS

S6202-F-73928 - Four Inch Square Enclosure - SBM Sheet Brass Assemblies for Electrical Fixtures and Fittings Type Z-4A.
S6502-74117 - Microphone Plug and Receptacle for Shipboard Announcing Equipment.
S6502-74164 - Microphone Magnetic-Type IC/MPM 3.
815-1853054 - Portable Ship Control Unit Submarine Bridge.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York).

3. REQUIREMENTS

3.1 Qualification. - The AN/WIC equipment furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 General features. - The equipment shall be in accordance with the following paragraphs of Specification MIL-I-983 in addition to the requirements specified herein: (Whenever a requirement of Specification MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

General information
Definitions
Materials, general
Stressed or vital parts
Substitution of (equal or superior) materials
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Aluminum
Magnesium
Iron or steel
Brass
Bronze
Copper
Copper-beryllium alloy
Copper-nickel alloy
Nickel-copper alloy (Monel)
Copper-nickel-zinc alloy (nickel silver and German silver)(nickel silver shall apply)
Zinc
Other metals
Springs
Plastics
Ceramics
Impregnating, filling, potting, embedding and encapsulating compounds
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Gaskets
Washers
Parts-electrical-general
Electron tube or capacitor sockets
Capacitors
Variable resistors
Transformers
Relays
Electrical tapes

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Switches
Indicator lights and lampholders
Fuses
Fuseholders and fuseclips
Enclosure - accessibility
Enclosure - degree of (dripproof)
Minimum sheet metal thicknesses (watertight enclosures)
Through bolting
Stuffing tubes
Ventilation
Size (for submarines)
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Welding
Temperature and humidity (special cold limit applies)
Shock, vibration and inclination
Primary power supply circuits
Power supply tolerances
Personnel protection
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Internal subassembly connection (test cable required)
Terminal boards connectors and terminals
Wiring
Color coding
Electrical insulation
Drawings general
Drawings - preliminary
Drawings - working
Schematic diagrams
Wiring diagram
Drawing list
Assembly drawings
Interchangeability and standardization
Technical manuals (systems manual type III required containing appropriate data required for type I manuals)
Repair parts (electronic)
Designation and marking (serial numbers are required)
Design for production
Reports
Nomenclature
Workmanship and general examination

3.2.1 Introduction. - The AN/WIC intercommunications system is a highly intelligible system designed to supply tactical interior communication needs for the efficient command and control of a submarine. The major design features provide for the following modes of operation:

(a) One-way broadcast of commands and announcements to select areas or to an entire ship.
(b) Group announcing systems for intercommunication between stations performing common or related functions.
(c) Two-way intercommunication between stations provided with selector switch facilities.
(d) Alarm signals which may be broadcast over the general announcing system.
(e) An emergency reporting system providing for the one-way communication of emergency reports from any compartment in the submarine to the control room.
(f) An escape trunk communication system which may be interconnected to selected stations such as the Conning station or Diving Officer station and which in an emergency may be disconnected from the ship's communication system and operating on its own auxiliary storage battery supply, may be used for communication between the escape trunk and adjacent compartments.
(g) A system for use in the Sound Room which provides for the selective break-in on a Sonar operator for purposes of communication. The Sonar operator is provided with a split headset; one earphone unit is permanently connected to the Sonar receiver and the other unit is normally connected to the Sonar receiver, which may be switched by a relay to receive messages from the sonar evaluator.
(h) A portable loudspeaker control unit which may be connected to the ship control unit extension cable for use in communicating from the deck watch to the control room when the ship is at anchor or in port. The portable unit may also be used on the general announcing system.

3.2.2 System components. - The components included in the AN/WIC intercommunication system are described briefly in the following subparagraphs:

3.2.2.1 Intercom Set Control C-3047/WIC. - This is an amplifier control (talk-back control box) and is the basic station in the system. This unit may be used either by itself for talk-back operation or used in conjunction with a control box for selective communication.

3.2.2.2 Intercom Set Control C-3048/WIC. - This amplifier control is similar to the amplifier control described in 3.2.2.1, except that it has a push-switch for direct report to the Conning station.

3.2.2.3 Intercom Set Control C-3223/WIC. - This amplifier control is similar to the amplifier control described in 3.2.2.1, except that it has a selector switch for selection of either of two circuits, or both.

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3.2.2.4 Intercom Set Control C-3224/WIC. - This amplifier control is similar to the amplifier control described in 3.2.2.1, except that it has a selector switch for selection of either of two circuits.

3.2.2.5 Intercom Set Control C-3051/WIC. - This is a 9-station control box and is used to provide communication on any circuit or between any station selected by one of the push-button selector switches. This unit also provides for monitoring of various circuits.

3.2.2.6 Intercom Set Control C-3054/WIC. - This is a 15-station control box and is similar in function and operation to the 9-station control box described in 3.2.2.5, except that it provides communication between a larger number of stations.

3.2.2.7 Intercom Set Control C-3052/WIC. - This is a 15-station control box with radio and is similar in function and operation to the 15-station control box described in 3.2.2.6, except for the inclusion of a handset receptacle for radiophone use and a volume control to set the radio monitor loudspeaker level.

3.2.2.8 Intercom Set Control C-3053/WIC. - This is a 15-station control box for Launcher and is similar in function and operation to the 15-station control box described in 3.2.2.6, except that it provides for missile compartment intraspace communication.

3.2.2.9 Intercom Set Control C-3050/WIC. - This is a 31 MC control box and is used in association with the escape trunks to provide communication.

3.2.2.10 Storage Battery BB-420/WIC. - This auxiliary battery supplies power for the 31 MC escape hatch system independent of the main power circuit.

3.2.2.11 Battery Box CY-2588/WIC. - The battery box houses the auxiliary battery (BB-420/WIC) and is mounted on the bulkhead in the vicinity of the 31 MC station.

3.2.2.12 Intercom Set Control C-3049/WIC. - This is a periscope/bridge control box and contains selector switches and relays to provide communication from the periscope and the bridge.

3.2.2.13 Amplifier Assembly AM-2318/WIC. - This is a dual 20-watt amplifier. One of the amplifiers is used to power the monitor loudspeaker on the bridge, and the other is used to power the talk/listen loudspeaker in the listen condition.

3.2.2.14 Intercom Set Control C-3056/WIC (for six operators). - This is a sonar break-in control box and is used in the sound room to control the communication with the sonar operators. This unit also provides facilities for monitoring the sonar signals.

3.2.2.15 Interconnecting Box J-1070/WIC. - This is a control box used in conjunction with the sonar break-in control box to provide communication for a sonar operator, and to provide for selecting of either the Conn Report or Torpedo Detection circuits.

3.2.2.16 Interconnecting Box J-1071/WIC. - This is a control box for a sonar operator. This unit is similar to the control box described in 3.2.2.15, except that it permits selecting only the Conn Report.

3.2.2.17 Electrical Equipment Cabinet CY-2671/WIC. - This is a centrally located control cabinet which provides circuit disconnect and termination facilities and contains the following components:

(a) Storage battery BB-415/WIC.
(b) Power supply PP-2221/WIC (two units).
(c) Power distribution panel SB-1056/WIC.
(d) Electrical test panel SB-991/EIC (contains two each of amplifier-oscillator group OA-2176(v)/WIC, amplifier assembly AM-2158A/WIC, and AF amplifier AM-2118/WIC).

3.2.2.18 Power Supply Failure Alarm BZ-55/WIC. - This is a power supply alarm signal which provides an audible and visual indication if power supply failure occurs.

3.2.2.19 Intercom Set Control C-3685/WIC. - This is a portable loudspeaker which is used at the Deck Watch station to provide for general announcements or communication with the Control Room while the ship is in port or at anchor. It is equipped with a connector for attaching the extension cable for the ship's control unit.

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3.2.2.20 Modification kit MK 518/WIC. - This is a unit for installation in amplifier-loudspeaker AM-2319/WIC or AM-2320/WIC which controls the volume of the loudspeaker between two pre-set levels.

3.2.2.21 Amplifier-loudspeaker AM-2319/WIC. - This is a low power loudspeaker comprising a 1-watt amplifier integrally mounted in a housing, in which is mounted a loudspeaker unit.

3.2.2.22 Amplifier-loudspeaker AM-2320/WIC. - This is a high power loudspeaker comprising a 20-watt amplifier integrally mounted in a housing, upon which is mounted a loudspeaker unit.

3.2.2.23 Interconnecting Box J-1073A/WIC. - This is a headset receptacle box which provides a junction in those areas where it is necessary to plug in a headset at a position remote from the amplifier control or at a position where it is desired to provide headsets for two operators, but only one station.

3.2.2.24 Interconnecting Box J-1072A/WIC. - This is a microphone receptacle box which provides relay switching between two microphones connected to it.

3.2.2.25 Switch Box SA-718/WIC. - This is an emergency reports switch and is a spring return device which permits the transfer of a sound powered telephone from its normal communication circuit to a special circuit for reporting to the Control Room and Maneuvering Room.

3.2.2.26 Dynamic Microphone Element M-99/U. - This is the basic noise cancelling microphone element used in the following acoustical units:

3.2.2.26.1 Handheld microphone M-98/U. - This is a dynamic noise cancelling microphone which is used in conjunction with an amplifier control to provide for communication in noisy areas.

3.2.2.26.2 Handheld microphone M-114/U. - This dynamic microphone is the same as the noise cancelling microphone described in 3.2.2.26, but is equipped with a standard plug in accordance with Drawing S6502-74117.

3.2.2.26.3 Handheld microphone M-106/U. - This dynamic microphone is the same as the noise cancelling microphone described in 3.2.2.26, but is equipped with a U-77/U plug in accordance with Specification MIL-C-10544.

3.2.2.26.4 Microphone holder. - This is a handheld microphone holder used to hold microphones M-98/U, M-106/U, M114/U and IC/MPM-6 in accordance with Drawing S6502-74164.

3.2.2.26.5 Headset-Microphone H-178/U. - This headset comprises a receiver and a noise cancelling microphone for use with an amplifier control for auixiliary means of communication in lieu of amplifier control loudspeaker.

3.2.2.26.6 Cord assembly electrical CX-4692/WIC. - This is an extension cord which is used to connect the headset-microphone H-179/U to a control box.

3.2.2.27.7 Cord assembly electrical CX-4691/WIC. - This is an extension cable which is used to connect the headset/microphone H-178/U to the amplifier control or junction box.

3.2.2.26.8 Headset-Microphone H-179/U. - This split circuit headset-microphone is used by the sonar operator to enable him to monitor the sonar and at the same time communicate with other stations.

3.2.2.26.9 Handset H-177/U. - This is a telephone type handset comprising a receiver and a noise cancelling microphone for use with an amplifier control to provide an auixiliary means of communication in lieu of amplifier control loudspeaker.

3.2.2.27 Distribution Box J-1066/WIC. - This is a type I distribution box which is used as a junction point between intercom station. It also provides for disconnecting local stations for maintenance.

3.2.2.28 Distribution Box J-1067/WIC. - This is a type II distribution box which is used as a junction point between local stations and common circuits to the electrical equipment cabinet. It also provides for disconnecting local stations for maintenance.

3.2.2.29 Distribution Box J-1068/WIC. - This is a type III distribution box and is similar to the type II distribution box, described in 3.2.2.28, except for arrangement of disconnecting plugs and terminal boards.

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3.2.2.30 Distribution Box J-1069/WIC. - This is a type IV distribution box and is similar to the type II distribution box described in 3.2.2.28, except for arrangement of disconnecting plugs and terminal boards.

3.2.3 Amplifiers and oscillators. - The following amplifiers and signal generators are used in the AN/WIC system.

3.2.3.1 AF Amplifier AM-2120A/WIC. - The preamplifier is designed specifically to amplify voice signals from the microphones and to apply the signal to the 1-watt amplifier.

3.2.3.2 AF Amplifier AM-2119A/WIC. - The 1-watt power amplifier is used to amplify the output of the preamplifier to the normal 1-watt audio line level or to power a low power loudspeaker. It is also used as an isolation amplifier and as an output amplifier for the signal generator.

3.2.3.3 Amplifier Assembly AM-2158A/WIC. - The line amplifier combines a preamplifier and a 1-watt amplifier into a plug-in assembly. It is used as both a talking and listening amplifier in the amplifier control, periscope/bridge control box, sonar break-in control box, and the 31 MC control box.

3.2.3.4 AF Amplifier AM-2118A/WIC. - The 20-watt power amplifier is used in the high-powered loudspeaker assembly, and in the dual amplifier assembly and 31 MC control box to power loudspeakers in the bridge ship control unit and escape trunk respectively.

3.2.3.5 Amplifier-Oscillator Group OA-2176(v)/WIC. - The alarm signal generator comprises an assembly containing plug-in modules which will produce various alarm signals.

3.2.4 Additional system components. - The components listed in table I are supplied by the installing activity, and are necessary for operation of the system.

Table I - Additional system components.

Quantity	Item	Nomenclature
2	Loudspeaker	LS-450/B
1	Portable ship control unit (for bridge)	Drawing 815-1853054
As required	Foot switch	----------
As required	Microphone	IC/MPM-6
As required	Microphone jackbox	IC/MJB
As required	Alarm contact makers	Types S-2JRM3

3.2.5 Mechanical and electrical construction requirements for each system component. - The mechanical construction requirements for each component shall be in accordance with the following paragraphs in addition to those specified herein. Wiring and terminal markings shall be in accordance with the applicable portions of figure 1 (see 3.2.12).

3.2.5.1 Intercom Set Control C-3047/WIC. -

3.2.5.1.1 Enclosure. - The enclosure shall consist of a two-section fabricated or drawn aluminum case. The front section shall contain all controls required for operation. Associated electrical and mechanical components shall be mounted on the rear of the front section. The rear section shall contain terminal boards for connection to ship's wiring. Mounting holes shall be provided for bulkhead mounting. Electrical connections between the front and rear section shall be made with plug and jack connectors. The plugs shall be mounted on the front section and the jacks in the rear section. Positive mechanical guides and guide pins shall be provided to align the plug and jack connectors and to support the front section when it is inserted and secured in its case. The front section shall be secured with captive thumb screws. Handles shall be provided to assist in insertion and withdrawal of the front section from the case. Cable entrances shall be provided at the top and bottom by means of removable plates. The plates shall be suitable for drilling and mounting of standard Navy terminal tubes. The assembled case shall be dripproof. Drainage shall be provided in the bottom.

3.2.5.1.2 Maximum weight. - The maximum weight shall be 12.5 pounds.

3.2.5.1.3 Maximum size. - The maximum size shall be: height 7.7 inches, width 9.5 inches, depth 8.3 inches.

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3.2.5.1.4 Mounting dimensions. - Four 0.375 inch diameter holes spaced on 7.75 inch by 5 inch mounting centers shall be provided through the rear of the case.

3.2.5.1.5 Controls. - The following controls and components shall be required on the front panels:

(a) Volume control.
(b) Indicator light.
(c) Press to talk switch.
(d) Headset receptacle and cap.
(e) Loudspeaker.
(f) Line amplifier module.

3.2.5.1.6 Headset-Microphone Jack. - The headset microphone jack shall be type U-79/U, Amphenol type No. 164-7U, or equal.

3.2.5.2 Intercom Set Control C-3048/WIC. - This amplifier control is identical to the amplifier control specified in 3.2.5.1, except that it has an additional momentary contact switch on the front panel for Conn Report.

3.2.5.3 Intercom Set Control C-3223/WIC. - This amplifier control is identical to the amplifier control specified in 3.2.5.1, except that it has an additional momentary selector switch for selection of either of two circuits, or both.

3.2.5.4 Intercom Set Control C-3224/WIC. - This amplifier is identical to the amplifier control specified in 3.2.5.1, except that it has an additional momentary selector switch for selection of either of two circuits.

3.2.5.5 Intercom Set Control C-3051/WIC. -

3.2.5.5.1 Enclosure. - The enclosure shall consist of a two section fabricated or drawn aluminum case. The front section shall contain all controls required for operation. Associated electrical and mechanical components shall be mounted on the rear of the front section. The rear section shall contain terminal boards for connection to ships wiring and mounting holes for bulkhead mounting. Electrical connections between the front and rear sections shall be made with plug and jack connectors. The plugs shall be mounted on the front section and jacks in the rear section. Positive mechanical guides and guide pins shall be provided to align the plug and jack connectors and to support the front section when it is inserted and secured in the case. The front section shall be secured with knurled thumb screws with slotted heads. Handles shall be provided to assist in insertion and withdrawal of the front section from the case. Cable entrances shall be provided at the top and the bottom by means of removable plates. The plates shall be suitable for drilling and mounting of standard Navy terminal tube. The assembled case shall be dripproof. Edge lighting techniques shall be used for illuminating functional and station designation. The illuminating panel shall be a black plastic material finished to reduce light reflection to a minimum. The panel shall be engraved to indicate functional controls. Station selector switches and the monitor switches shall be provided with blank spaces for engraving of station designation by the installing activity. The engraved designations shall be white and shall appear red when illuminated. Panel lamps and illumination panel shall be removable from the front with a minimum of disassembly. A continuously variable dimmer control shall be provided to permit control of panel illumination in darkened to average lighted areas.

3.2.5.5.2 Maximum weight. - The maximum weight shall be 7.5 pounds.

3.2.5.5.3 Maximum size. - The maximum size shall be: height - 5.5 inches, width 9.6 inches, depth 8.3 inches.

3.2.5.5.4 Mounting dimensions. - Four 0.375 inch diameter holes spaced on 7.75 inch by 2.8 inch mounting centers shall be provided through the rear of the case.

3.2.5.5.5 Controls. - The following controls shall be required on the front panel:

(a) Illumination panel dimmer control
(b) Monitor volume control.
(c) Monitor switches.
(d) Station selector switches.
(e) Station standby switch.

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3.2.5.6 Intercom Set Control C-3054/WIC. - The requirements of 3.2.5.5 apply, except as follows:

3.2.5.6.1 Maximum weight. - The maximum weight shall be 10.4 pounds.

3.2.5.6.2 Maximum size. - The maximum size shall be, height 7.5 inches, width 9.5 inches, depth 8.3 inches.

3.2.5.6.3 Mounting dimensions. - Four 0.375 inch diameter holes spaced on 7.75 inch by 5 inch mounting centers shall be provided.

3.2.5.7 Intercom Set Control C-3052/WIC. - The requirements of 3.2.5.5 apply, except as follows:

3.2.5.7.1 Maximum weight. - The maximum weight shall be 10.6 pounds.

3.2.5.7.2 Controls. - The following additional controls shall be provided on the front panel:

(a) Radio handset receptacle.
(b) Radio monitor volume control.
(c) Carrier on indicator lamp.

3.2.5.8 Intercom Set Control C-3053/WIC. - The requirements of 3.2.5.6 apply.

3.2.5.9 Intercom Set Control C-3050/WIC. - The requirements of 3.2.5.5 apply, except as follows:

3.2.5.9.1 Panel illumination. - Not applicable.

3.2.5.9.2 Maximum weight. - The maximum weight shall be 11.2 pounds.

3.2.5.9.3 Maximum size. - The maximum size shall be: height 7.7 inches, width 9.5 inches, depth 10.3 inches.

3.2.5.9.4 Controls. - The following controls shall be provided on the front panel:

(a) Selector switches.
(b) Test jacks.
(c) Volume control.
(d) Line amplifier.
(e) Fuseholder.

3.2.5.9.5 AF Amplifier AM-2118A/WIC. - The amplifier shall be mounted in the chassis to drive the escape trunk loudspeaker and line amplifier assembly AM2158A/WIC for use as both talking and listening, shall be mounted in the enclosures.

3.2.5.10 Storage Battery BB-420/WIC. - The storage battery shall be a 24 volt, 3 ampere hour sintered plate nickel cadmium type, Sonatone number 19205, or equal.

3.2.5.11 Battery Box CY-2588/WIC. - This shall be a rectangular metal box to house storage battery BB-420/WIC. It shall be designed for bulkhead mounting and shall have a removable cover to permit easy access to the battery. Holes or louvres shall be provided in the case for ventilation. The battery shall be positively secured in the box and readily removable for service.

3.2.5.11.1 Maximum weight. - The maximum weight shall be 18 pounds.

3.2.5.11.2 Maximum size. - The maximum size shall be: height 7.00 inches, width 11.2 inches, depth 6.5 inches.

3.2.5.11.3 Mounting. - Three mounting holes 0.437 inch in diameter shall be provided for bulkhead mounting. Two holes shall be located at the top and one at the bottom. The top mounting holes shall be spaced on 11-inch centers, and the bottom hole shall be located at the center 4.5 inches below the top holes.

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3.2.5.12 Intercom Set Control C-3049/WIC. -

3.2.5.12.1 Enclosure. - The enclosure shall be designed for bulkhead mounting. The front panel shall be hinged and shall be supported when in the open position. Captive thumb screws shall be used to secure the front panel to the case. Panel illumination shall be provided in accordance with the method specified in 3.2.5.5. Terminal boards for connection to ships wiring shall be provided in the rear of the case. The enclosure shall be dripproof.

3.2.5.12.2 Maximum weight. - The maximum weight shall be 30 pounds.

3.2.5.12.3 Maximum dimensions. - The maximum dimensions shall be: height 16.9 inches, width 12.5 inches, depth 10.2 inches.

3.2.5.12.4 Mounting dimensions. - Four 0.375 inch diameter holes spaced on 7.75 by 5 inch mounting centers shall be provided through the rear of the case.

3.2.5.12.5 Controls. - The following controls shall be provided on the front panel. Topside controls shall be grouped at the top of the panel and Periscope station controls at the bottom.

Topside controls

(a) Bridge microphone selector switch.
(b) Bridge circuit "ON" indicator.
(c) Bridge circuit "Busy" indicator.
(d) Topside "ON" "OFF" "INPORT" selector switch.
(e) Bridge monitor volume control.
(f) Bridge monitor selector switches.
(g) Bridge listen volume control and panel lamp.

Periscope controls

(a) Monitor selector switches.
(b) Pushbutton selector switches.
(c) Radio loudspeaker volume control.
(d) Monitor loudspeaker volume.
(e) Radio carrier on indicator.
(f) Radio handset receptacle.
(g) Panel illumination dimmer control.
(h) Panel lamps.
(i) Line amplifier.

3.2.5.13 Amplifier Assembly AM-2318/WIC. - Enclosure design, size and mounting dimensions shall be in accordance with 3.2.5.1.1, except as follows:

3.2.5.13.1 Maximum weight. - The maximum weight shall be 17.0 pounds.

3.2.5.13.2 Maximum size. - The maximum size shall be: height 7.5 inches, width 9.5 inches, depth 10.3 inches.

3.2.5.13.3 Mounting dimensions. - Four 0.375 inch diameter holes spaced on 7.75 inches by 5 inch mounting centers shall be provided.

3.2.5.13.4 Two AM-2118A/WIC 20-watt amplifiers shall be mounted on the rear of the front panel assembly.

3.2.5.14 Intercom Set Control C-3056/WIC. - Enclosure design, weight, size and mounting dimensions shall be in accordance with 3.2.5.13, 3.2.5.13.2, and 3.2.5.13.3.

3.2.5.14.1 Maximum weight. - The maximum weight shall be 14 pounds.

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3.2.5.14.2 Controls. - The following controls and components shall be provided on the front panel:

(a) Break-in indicator lamp.
(b) Operator break-in switches.
(c) Sonar monitor switch(s).
(d) Monitor volume control.
(e) Line amplifier.
(f) Torpedo detection switch.
(g) Conning station call switch.
(h) Operating mode selection switch.
(i) Operating mode indicator lamp.

3.2.5.15 Interconnecting Box J-1111/WIC. -

3.2.5.15.1 Enclosure. - Enclosure shall consist of a dripproof box assembly in accordance with the following weight and dimensions: weight 1.2 pounds, height 7.75 inches, width 2.18 inches, depth 4.13 inches.

3.2.5.15.2 Controls. - The following controls shall be provided on the cover assembly:

(a) Torpedo detection switch.
(b) Conning station call switch.
(c) Busy indicator lamp.
(d) Volume control.

3.2.5.16 Interconnecting Box J-1112/WIC. - This box shall be identical to J-1111/WIC specified in 3.2.5.15, except that the torpedo detection switch is not required.

3.2.5.17 Electrical Equipment Cabinet CY-2671/WIC-2. -

3.2.5.17.1 Cabinet. - The cabinet shall be of formed sheet metal construction with internal supporting frame work. It shall be designed for deck mounting with provision for sway bracing at the top. Cable entrances shall be provided at the top and bottom by means of removable plates of sufficient thickness to mount standard Navy terminal tubes. The cabinet shall be dripproof. The front panels of the cabinet shall be functionally arranged in descending order from the top as follows:

(a) Switch, fuse, and relay panel.
(b) Electrical test, oscillator, and amplifier stowage panel.
(c) Power distribution panel.
(d) Power supplies (2).
(e) Battery compartment.

Louvers shall be provided in cabinet to provide natural ventilation of the battery compartment and in other areas as necessary to provide adequate dissipation of heat from the cabinet. Terminal boards for connection to ship's wiring shall be mounted in the rear of the cabinet. Access to the terminal boards shall be from the front.

3.2.17.1.1 Maximum dimensions. - The maximum dimensions shall be: height 70 inches, width 21 inches, depth 15 inches.

3.2.5.17.1.2 Maximum weight. - The weight shall not exceed 485 pounds.

3.2.5.17.2 Switch, fuse, and relay panel. - This panel shall be hinged and provided with a folding locking type brace to hold the panel in the open position. The panel shall be secured in the closed position by means of captive thumb screws. Switches, fuses, and designation markers shall be grouped in accordance with function on the front panel. The grouping in descending order from the top shall be as follows:

(a) Station disconnect switches (16); fuses, and designation strips.
(b) Emergency reports disconnect switches (20) and designation strips.
(c) General announcing group disconnect switches (30) and fuses.

Amplifiers, relays, and line loading, resistors shall be mounted on the rear of the hinged panel.

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3.2.5.17.3 Electrical test Panel SB-991/WIC. - This shall be a hinged panel with a means of supporting the panel when the panel is in the open position. The panel shall be secured in the closed position by means of captive thumb screws. Direct access to ships terminal boards shall be provided when the panel is in the open position.

3.2.5.17.3.1 Controls. - The following controls and components shall be provided on the hinged panel:

(a) Two alarm signal generator modules.
(b) Two line amplifiers modules.
(c) One 20 watt amplifier module.
(d) Test selector switch for testing amplifiers and signal generators.
(e) Test meter.
(f) Relay test socket.
(g) Relay test indicator lamp.
(h) Relay test switch.
(i) Test panel power on switch and fuse.
(j) Test panel power on indicator lamp.
(k) Signal generator selector switch.
( ) Alarm test switches.
(m) Alarm test indicator lamps.
(n) Headset receptacle.

All indicator lamps shall be removable from the front. Amplifier and signal generator modules shall be removable from the front. One of the line amplifiers is extra and requires stowage facilities only.

3.2.5.17.3.2 Test meter. - This shall be a DB meter scale arrangement - 10/0/+6DB meeting the requirements of Specification MIL-M-10304.

3.2.5.17.3.3 Alarm Signal Generator Module OA-2176(V)/WIC-1. -The module shall contain facilities for seven plug-in alarm signal generator assemblies. Each assembly shall contain all the necessary components to generate the alarms specified independent of one another. Facilities shall be provided to prevent accidental insertion of assemblies into the wrong priority position. The 1-watt Amplifier AM-2119A/WIC shall be mounted on the rear of the module. The complete module shall be a plug-in assembly secured to the test panel by means of captive thumb screws. Priority of the signal shall be determined by the position of signal generator assemblies in the module. The number, type, and priority will be specified in the order.

3.2.5.17.4 Power distribution panel SB-1056/WIC. - The panel and chassis shall be a drawer type assembly removable from the front. Handles shall be provided on the front panel for removing the assembly from the cabinet. Captive thumb screws shall be provided to secure the assembly in the cabinet. The panel shall contain the system power distribution relays, fuses, power supply switching and test facilities.

3.2.5.17.4.1 Controls. - The following controls shall be provided on the chassis and panel:

(a) Power supply selector switch.
(b) Power supply "Power On" indicator lamps.
(c) 110 volt a.c. power fuses.
(d) 28 volt d.c. power fuses.
(e) Blown fuse indicator lamps.
(f) System test switch.
(g) Standby power supply "Power On" switch.
(h) Standby power supply test switch.
(i) Signal reset switch.
(j) Power supply power failure indicator lamp.
(k) Voltmeter and ammeter.

3.2.5.17.5 System Power Supply. - The system power supply shall consist of two complete power supplies, PP-2221/WIC. Each power supply shall consist of a full wave bridge type silicon diode rectifier to convert 115 volt, 60 cycle to 28 volt dc (see 3.2.7). All control circuits and amplifiers shall operate at this voltage. Each power supply shall be mounted on a panel-chassis assembly which can be removed from the panel for servicing. A hinged front panel secured by captive thumb screws shall be provided to permit quick access to power supply test points and silicon diodes. Handles shall be provided to withdraw the power supply from the cabinet.

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3.2.5.17.5.1 System stowage battery BB-415/WIC. - This shall be a 24 volt sintered plate nickel-cadmium storage battery Sonotone Corp. No. 19174, or equal. The battery shall be securely mounted on slides to permit removal for servicing. Identical provisions shall be made for a second storage battery. Only one battery will be supplied with the equipment unless otherwise specified in the contract or order.

3.2.5.18 Power supply failure alarm BZ-55/WIC. -

3.2.5.18.1 Enclosure. - The enclosure shall consist of a box assembly and cover assembly in accordance with Drawing S6202-F-73928.

3.2.5.18.2 Controls. - The following controls shall be provided on the cover assembly:

(a) Momentary switch.
(b) Power failure indicator lamp.

3.2.5.19 Intercom Set Control C-3685/WIC. -

3.2.5.19.1 Enclosure. - The enclosure shall consist of a portable case, transformers, resistors and terminal board and connector (connector to be compatible with Drawing 815-1853054). The case and cover is hinged at the bottom with locking catches on each side. The case cover shall contain a talk/listen loudspeaker mounted on the front. The top of the cover shall contain all the necessary controls for selecting the mode of operations and carrying handle. Selection mode shall be compatible with that shown on Drawing 815-1853054 (for MC circuits).

3.2.5.19.2 Maximum weight. - The maximum weight shall be 14.5 pounds.

3.2.5.19.3 Maximum dimensions. - The maximum dimensions shall be: height 13.3 inches, width 14.5 inches, depth 7.3 inches.

3.2.5.19.4 Controls. - The following controls and components will be required on the case cover:

(a) Volume control.
(b) Control room call switch.
(c) 1 MC call switch
(d) Headset-microphone receptacle and cap.
(e) Loudspeaker talk/listen.

3.2.5.20 Modification Kit MK-518/WIC. - This is a unit to be used to modify an AM-2319/WIC or AM-2320/WIC loudspeaker to provide control of the volume between two preset levels. It shall be mounted within the amplifier-loudspeaker case.

3.2.5.20.1 This unit is a bracket which contains a relay, volume control, and resistor. It shall be mounted to the back-plate of an amplifier-loudspeaker with screws. Connections shall be made by attaching the wires to the terminal board of the amplifier-loudspeaker.

3.2.5.20.2 Maximum dimensions. - The maximum dimensions shall be: height 3 inches, width 2 inches, depth 2 inches.

3.2.5.21 Amplifier loudspeaker AM-2319/WIC. -

3.2.5.21.1 Enclosure. - The enclosure shall consist of a case and cover of drawn or fabricated aluminum. The case shall be designed for bulkhead mounting and shall be provided with cable entrance plates top and bottom. The case shall contain a terminal board, for connection to ships wiring. The cover shall be secured to the case with a hinge and four captive thumbscrews. A low-power loudspeaker shall be located on the cover front and the amplifier and volume control mounted on the inside of the cover.

3.2.5.21.2 Maximum weight. - The maximum weight shall be 6.5 pounds.

3.2.5.21.3 Maximum dimensions. - The maximum dimensions shall be: height 10.2 inches, width 7 inches, depth 5.4 inches.

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3.2.5.21.4 Mounting dimensions. - Four 0.375 inch diameter mounting holes spaced on 7.75 by 5 inch mounting centers shall be provided through the rear of the case.

3.2.5.21.5 Controls. - A volume control shall be provided on the inside of the front cover.

3.2.5.22 Amplifier Loudspeaker AM-2320/WIC. -

3.2.5.22.1 Enclosure. - The enclosure shall consist of a case and cover of drawn or fabricated aluminum. The case shall be designed for bulkhead mounting and shall be provided with cable entrance plates, top and bottom. The inside rear of the case shall contain a terminal board for connection to ships wiring and a connector which shall be mated with the connector on the amplifier. The cover shall be secured to the case with a hinge and four captive thumbscrews. A high-power horn type loudspeaker shall be located on the outside of the cover front.

3.2.5.22.2 Maximum weight. - The maximum weight shall be 13.3 pounds.

3.2.5.22.3 Maximum dimensions. - The maximum dimensions shall be: height 10.2 inches, width 7 inches, depth 9.9 inches.

3.2.5.22.4 Mounting dimensions. - Four 0.375 inch diameter holes spaced on 7.75 by 5 inch mounting centers shall be provided through the rear of the case.

3.2.5.23 Interconnecting Box J-1073A/WIC. -

3.2.5.23.1 Enclosure. - The enclosure shall consist of a box assembly and cover assembly in accordance with Drawing S6202-F-73928. The box assembly shall contain terminal boards and two relays. The cover shall contain a headset receptacle and cap.

3.2.5.24 Interconnecting Box J-1072A/WIC. -

3.2.5.24.1 Enclosure. - The enclosure shall consist of a box assembly and a cover assembly in accordance with Drawing S6202-F-73928. The box assembly shall contain terminal boards and one relay. The cover shall contain a microphone receptacle and cap.

3.2.5.25 Switch box (emergency reports) SA-718/WIC. -

3.2.5.25.1 Enclosure. - The enclosure shall consist of a box assembly and a cover assembly in accordance with Drawing S6205-F-73928. The box assembly shall contain a terminal board. The cover shall contain a spring return switch with a locking device in the return position. Weight and dimensions shall be as follows: weight 3.5 pounds; Dimensions - height 5.9 inches, width, 5.9 inches, depth 3.3 inches.

3.2.5.26 Dynamic microphone element M-99/U. -

3.2.5.26.1 Enclosure. - The enclosure shall consist of a black nylon molded case for the microphone element and transformer. The microphone element shall be designed for use with headsets H-179/U and H-178/U, handset H-177/U and handheld microphones M-98/U, M-97/U, M-114/U and M-106/U. The microphone element shall be a noise-cancelling type for use in high noise level areas where high intelligibility is required.

3.2.5.26.2 Maximum weight. - The maximum weight shall be: 21 grams complete; 14 grams (excluding transformer).

3.2.5.26.3 Maximum dimensions. - The maximum dimensions shall be: height 1.95 inches, width 1.063 inches, depth 0.58 inches.

3.2.5.26.4 Mounting dimensions. - The microphone element shall be provided with two 2-56 screws located on the back of case for mounting to headsets, handsets and handheld microphones.

3.2.5.27 Handheld microphone M-98/U. -

3.2.5.27.1 Microphone. - The handheld microphone shall consist of a microphone element (M-99/U, see 3.2.5.26), a black molded thermo plastic case and cover, a press to talk switch and a retractable - non-kinking connecting cable with terminal lugs. The case shall consist of a press to talk switch located on the

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top, a terminal block on the inside and a retractable connecting cable on the bottom. The back of the case shall be fitted with a latch plate for securing the microphone to a holder (for delineation of the microphone holder see Drawing S6502-74164). The cover shall contain the microphone element M-99/U and a terminal block.

3.2.5.27.2 Maximum weight. - The maximum weight shall be 0.80 pound.

3.2.5.27.3 Maximum dimensions. - The maximum dimensions shall be: height 4.25 inches, width 2 inches, depth 1.90 inches.

3.2.5.28 Handheld microphone M-97/U. -

3.2.5.28.1 Microphone. - The handheld microphone shall be identical to the M-98/U specified in 3.2.5.27, except that the microphone retractable connecting cable is equipped with a special pressure prrof plug (for delineation of plug, see Drawing S6501-73462).

3.2.5.29 Handheld microphone M-106/U. -

3.2.5.29.1 Microphone. - The handheld microphone shall be identical to the M-98/U specified in 3.2.5.27, except that the microphone shall be equipped with a U-77/U connector in accordance with Specification MIL-C-10544.

3.2.5.30 Handheld microphone M-114/U. -

3.2.5.30.1 Microphone. - The handheld microphone shall be identical to the M-98/U specified 3.2.5.27, except that the microphone shall be equipped with an H-29B plug in accordance with Drawing S6502-74117.

3.2.5.31 Microphone - holder assembly. -

3.2.5.31.1 Microphone holder. - The microphone holder shall be in accordance with Drawing S6502-74164.

3.2.5.32 Headset - Microphone H-178/U. -

3.2.5.32.1 Headset. - The headset shall consist of two receivers, a microphone element (M-99/U see 3.2.5.26), a headband, a microphone boom, and a combinedmicrophone-receiver cord assembly. The two receivers shall be mounted in hard noise excluding ear shells. The ear shells shall be attached to an adjustable headband. The two receivers shall be mounted in the ear shells and connected in parallel by a cord assembly attached to the headband. The cord assembly shall be provided with connector assemblies at each end for connection to the top of the receiver ear shells. The microphone element shall be mounted on an adjustable boom attached to the left ear shells by a thumbscrew. The combined microphone-receiver cord assembly shall be fitted with connector assemblies for connection at bottom of the left ear shell and the microphone element and shall terminate in the headset - microphone plug.

3.2.5.32.2 Maximum weight. - The maximum weight shall be 0.76 pound.

3.2.5.32.3 Maximum dimensions. - The maximum dimensions shall be: height 9.04 inches, width 7.60 inches, depth 3.50 inches.

3.2.5.33 Cord Assembly Electrical CX-4691/WIC. -

3.2.5.33.1 Cord assembly. - The cord assembly shall consist of a press-to-talk switch and connector assembly (U-94A/U) and a 12-foot five-conductor shielded cable terminated in a U-77/U connector in accordance with Specification MIL-C-10544.

3.2.5.33.2 Maximum weight. - The maximum weight shall be 0.69 pound.

3.2.5.34 Cord Assembly Electrical CX-4692/WIC. -

3.2.5.34.1 Cord assembly. - The cord assembly shall consist of connector assembly to accept plugs of H-179/U) and 6 feet of five-conductor shielded cable with terminal lugs.

3.2.5.34.2 Maximum weight. - The maximum weight shall be 0.34 pound.

3.2.5.35 Headset - Microphone H-179/U. -

3.2.5.35.1 Headset. - The headset shall be similar to the H-178/U specified in 3.2.5.32, except that the two receivers are not connected in parallel. The right and left receivers and the microphone shall be brought out separately and terminate in the headset microphone plug.

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3.2.5.35.2 Maximum weight. - The maximum weight shall be 0.80 pound.

3.2.5.36 Handset H-177/U. -

3.2.5.36.1 Handset. - The handset shall consist of a microphone element (M-99/U see 3.2.5.26), a receiver, handset shell, 5-conductor retractable cord (4 feet) and a press-to-talk switch.

3.2.5.36.2 Maximum weight. - The maximum weight shall be 1.15 pounds.

3.2.5.36.3 Maximum dimensions. - The maximum dimensions shall be: height 9 inches, width 2.50 inches, depth 3.36 inches.

3.2.5.37 Distribution Box J-1066/WIC. -

3.2.5.37.1 Enclosure. - The enclosure shall consist of a box assembly and a cover assembly of sheet brass. The distribution box shall be designed for bulkhead mounting with a cover secured to the box by a hinge and three thumbscrews. The box shall contain terminal boards for connecting to ship's wiring and connector assemblies for disconnecting local intercom station.

3.2.5.37.2 Maximum weight. - The maximum weight shall be 18.5 pounds.

3.2.5.37.3 Maximum dimensions. - The maximum dimensions shall be: height 10.3 inches, width 14.2 inches, depth 4.2 inches.

3.2.5.37.4 Mounting dimensions. - Six 0.531 inch diameter mounting holes spaced on 8.50 by 5.00 inch centers shall be provided through the rear of the box.

3.2.5.38 Distribution Box J-1067/WIC. -

3.2.5.38.1 Enclosure. - The enclosure shall be identical to the one specified in 3.2.5.37.1, except for connector assemblies for disconnecting local stations.

3.2.5.38.2 Weight and mounting dimensions shall be identical to those specified in 3.2.5.37.2 and 3.2.5.37.4.

3.2.5.39 Distribution Box J-1068/WIC. -

3.2.5.39.1 Enclosure. - The enclosure shall consist of a box assembly and a cover assembly of sheet brass. The distribution box shall be designed for bulkhead mounting with a cover secured to the box by a hinge and three thumbscrews. The box shall contain terminal boards for connecting to ship's wiring and connector assemblies for disconnecting local stations from the system.

3.2.5.39.2 Maximum weight. - The maximum weight shall be 21.5 pounds.

3.2.5.39.3 Maximum dimensions. - The maximum dimensions shall be: height 12.3 inches, width 14.3 inches, depth 4.2. inches.

3.2.5.39.4 Mounting dimensions. - Six 0.531 inch diameter mounting holes spaced on 1.05 by 5.5 inch centers shall be provided through the rear of the box.

3.2.5.40 Distribution Box J-1069/WIC. -

3.2.5.40.1 Enclosure. - The enclosure shall be identical to the one specified in 3.2.5.39.1, except for connector assemblies for disconnecting local stations.

3.2.5.40.2 Weight and mounting dimensions shall be identical to those specified in 3.2.5.39.2 and 3.2.5.39.4.

3.2.6 General requirements - mechanical and electrical. -

3.2.6.1 Toggle switches. - Toggle switches shall be in accordance with Specification MIL-S-3950. Rubber boots, in accordance with Specification MIL-B-5423, shall be provided on all toggle switches. Rubber boots will not be required on switches which have integral water proofing.

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3.2.6.2 Rotary switches. - Rotary switches shall meet the requirements of Specification MIL-S-3786. (Qualification is not required).

3.2.6.3 Push switches. - Push switches shall be in accordance with Specification MIL-S-6743 or MIL-S-8805.

3.2.6.4 Pushbutton switch assemblies, multiple selectors. - Pushbutton switch assemblies shall be mechanically interlocked so that only one button can be depressed at a time. Minimum spacing between pushbutton centers shall be 5/8 inch. Pushbuttons shall be plastic with metallic inserts. The switch shall be capable of a minimum of 100,000 operations without mechanical or electrical failure. This requirement applies with electrical loads equivalent to actual circuit loads connected through the switch contacts. Boots shall be provided on pushbutton shafts to prevent entry of dust and water into enclosures. Pushbutton switch assemblies shall meet the requirements of Specification MIL-S-901, except that the release of mechanically interlocked pushbutton switches during the shock test shall not be considered as a failure; however, the pushbutton selector switches shall not release when subjected to two blows of 800 inches-pounds in each of three axes as specified in Specification MIL-S-901.

3.2.6.5 Fuses. - Fuses shall be in accordance with Specification MIL-F-15160. One hundred fifteen volt a.c., 60 cycle supply shall be fused in both sides of the line. Twenty-eight volt d.c. circuits associated with station group action cutout switches in the electrical equipment cabinet (see figure 1) shall be fused to protect the station equipment complement shown on overall system schematic guidance diagrams.

3.2.6.6 Relays. - Relays except power-type contactors shall be plug-in type meeting the electrical and mechanical requirements of Specifications MIL-R-5757 or MIL-R-2033. The contacts of all relays for audio circuits shall be of the bifurcated type. Each contact point shall be capable of carrying the total load current of the circuit being switched. Two spare plug-in relays shall be mounted in the control equipment cabinet. Plug-in relay coil power consumption at 28 v. d.c. shall not exceed 48 milliamperes. Relays shall operate satisfactorily on a minimum of 18 volts. Power type contactors shall be in accordance with Specification MIL-R-2033. Contacts of power relays shall be capable of carrying 15 amps, 28 v. d.c.

3.2.6.7 Power supply rectifiers. - Power supply rectifiers shall be silicon diodes in accordance with Standard MIL-STD-701.

3.2.6.8 Semiconductor devices. - Amplifiers and oscillators shall employ transistors and other semiconductor devices exclusively. Transistors and diodes shall be in accordance with Specification MIL-E-1 and MIL-S-19500. Transistors shall be selected in accordance with Standard MIL-STD-701.

3.2.6.9 Printed circuits. - All amplifiers and oscillators shall be constructed as plug-in units using printed circuit board assemblies. Printed circuit boards shall be in accordance with Standard MIL-STD-275.

3.2.6.10 Encapsulation. - Amplifiers may be conformal coated provided they are repairable aboard ship.

3.2.6.11 Transformers. - Transformers shall be grade 4 or 5, class R, life expectancy X in accordance with Specification MIL-T-27. Transformers to be assembled to printed circuit boards and encapsulated may be grade 3.

3.2.6.12 Panel illumination and indicator lamps. - All 28 v. d.c. lamps shall be in accordance with Standard MS25237-327. Six volt direct current lamps shall be in accordance with Standard MS15544-1.

3.2.6.13 Capacitors. - Tantalum capacitors shall be used in the power supply. All other electolytic capacitors used in the design of this equipment shall meet the requirements of Specification MIL-C-3965, except for size and rating. Paper capacitors shall be in accordance with Specification MIL-C-25.

3.2.6.14 Volume controls. - Volume controls shall be in accordance with Specification MIL-R-94. The volume control shall provide a minimum control range of 40 decibels (db). At one-half rotation of the control, amplifiers shall provide rated output at the maximum position of the control. The gain shall be increased by 13 to 16 db.

3.2.6.14.1 Loud speaker volume control. - Loud speaker volume controls shall be screw driver adjustment type with locking nut.

3.2.6.15 Cross talk. - Cross talk between any channels during the time equipment is energized for operation shall not exceed minus 50 db. below the maximum listening level at any station equipment.

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3.2.6.16 Switching transients and oscillations. - Audible noise or oscillations in loudspeakers due to switching transients shall be not greater than 40 db below rated amplifier output.

3.2.7 System power supply electrical characteristics. - Each of the two rectified power supplied shall meet the following requirements with the input line voltage set at 115 volts, 60 cycles.

3.2.7.1 No load output voltage. - Output voltage shall be variable between 0 and 30 v. d.c. This shall be accomplished by means of a variable auto transformer in the 115-volt, 60 cycle input circuit.

3.2.7.2 Output voltage regulation. - With the output voltage control set to deliver 29 volts d.c. at the output terminals at no load conditions, the voltage shall not exceed 29 volts at loads from 0 to 6 amperes At loads from 6 to 12 amperes, the maximum voltage limit shall decrease linearly from 29 volts at 6 amperes to 24 volts at 12 amperes. At loads from 0 to 1 ampere, the minimum voltage limit shall decrease linearly from 29 volts at 0 ampere to 26.5 volts at 1 ampere, and from 1 to 12 amperes shall decrease linearly 26.5 volts at 1 ampere to 24 volts at 6 amperes to 21 volts at 12 amperes.

3.2.7.3 Ripple voltage. - Ripple voltage at the direct current supply output terminals shall not exceed 0.7 volt r.m.s. between no load and 13 ampere load, with power supply battery disconnected.

3.2.7.4 Input power consumption. - The input power consumption measured in the 115-volt, 60-cycle input line shall not exceed 460 watts, when the direct current output load current is 13 amperes.

3.2.8 Amplifiers. -

3.2.8.1 Frequency response. - When delivering rated power, all amplifiers shall have a frequency response which falls within the limits of +2 -3 db. between 200 and 6000 c.p.s.

3.2.8.2 Distortion at rated power out into rated load. - The harmonic distortion at 200, 400, 1000, 4000 and 6000 cycles per second shall not exceed 5 percent. With the power output reduced by 3 db by reducing the input level, the harmonic distortion at 100 cycles shall not exceed 5 percent at 1000 c.p.s.

3.2.8.3 Rated power output. - When connected to a resistive load of a value to draw rated power and with an input signal applied as specified in 4.5.7, the output at 100 cycles shall not fall below 3 db from the rated output.

3.2.8.4 Specific amplifiers. - Amplifiers used in this system shall be of three basic types, each type to be of plug-in modular construction. The three types will be: pre-amplifier, a 1-watt amplifier and a 20-watt amplifier. All amplifiers shall employ push-pull transformer coupled transistor circuitry.

3.2.8.4.1 Amplifier characteristics. - Amplifier characteristics shall be as follows:

3.2.8.4.1.1 1-Watt Amplifier AM-2119A/WIC. - Terminal designations for the AM-2119A/WIC shall be as follows:

Terminal A Input high
B Isolation Transformer input high
C Isolation Transformer output high
D Audio Common
E 8 ohm output high
F 150 ohm output high
G NC
H Ground
J Ground
K +28 volt direct current
L 150 ohm output low

3.2.8.4.1.1.1 Input impedance. - The input impedance at 1000 c.p.s. including the external isolation resistor shall be at least 27K ohms.

3.2.8.4.1.1.2 Output impedance. - The amplifier shall meet the requirements of this specification when the 8-ohm winding is operating into an 8-ohm resistive load or when the 150-ohm winding is loaded by a 150-ohm resistive load.

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3.2.8.4.1.1.3 Direct current power input. - The amplifier shall draw no more than 0.160 ampere at 28 v. d.c. when it is producing 1-watt signal output at 1000 c.p.s. at 25°C.

3.2.8.4.1.1.4 Gain. - The gain of the amplifier shall be such that when the amplifier is producing 1-watt output at 1000 c.p.s., the input signal voltage into the pad shown on figure 2 shall be between 12.0 and 24.0 volts.

3.2.8.4.1.1.5 Distortion. - At 1-watt output, the harmonic distortion from 200 c.p.s. to 6K c.p.s. shall not exceed 5 percent. At 0.5-watt output, the harmonic distortion from 100 to 199 c.p.s. shall not exceed 5 percent.

3.2.8.4.1.1.6 Frequency response. - Using 1 watt output at 1000 c.p.s. as a 0 db reference level, the frequency response at a constant signal voltage input shall be within ± 2 db from 200 to 6K c.p.s. The response from 100 to 199 c.p.s. shall be within +2 to -3 db.

3.2.8.4.1.2 20 Watt Amplifier AM-2118A/WIC. - Terminal designations for the AM-2118A/WIC shall be as follows:

Terminal	1	150-ohm direct input high
	2	150-ohm input to volume control moving arm
	3	NC
	4	Ground
	5	+28 v. d.c.
	6	Output low
	7	8 ohm output
	8	Volume control winding (jumpered to terminal 1 when volume control is used)
	9	Audio common input
	10	NC
	11	Ground
	12	+28 v. d.c.
	13	16-ohm output
	14	250-ohm output

3.2.8.4.1.2.1 Input impedance. - The input impedance of 1000 c.p.s. including the external isolation resistor shall be at least 22K ohms.

3.2.8.4.1.2.2 Output impedance. - The amplifier shall conform to the requirements of this specification when the 8-ohm winding is loaded with an 8-ohm resistive load. The amplifier shall also conform to the requirements of this specification when a 16-ohm resistive load is placed across the 16-ohm winding or a 250-ohm load is placed across the 250-ohm winding. The loads shall not be applied simultaneously.

3.2.8.4.1.2.3 Direct current power input. - The amplifier shall draw no more than 2.0 amperes at 28 v. d.c. when it is producing 20 watts signal output at 1000 c.p.s. at 25°C.

3.2.8.4.1.2.4 Gain. - A 12.25 v. r.m.s. signal input at 1000 c.p.s. through a 22K-ohm connected to terminal 2 shall produce a minimum of 20 watts output.

3.2.8.4.1.2.5 Distortion. - At 20 watts output, the harmonic distortion from 200 c.p.s. to 6K c.p.s. shall not exceed 5 percent. At 10 watts output, the harmonic distortion from 100 to 199 c.p.s. shall not exceed 5 percent.

3.2.8.4.1.2.6 Frequency response. - Using 20 watt output at 1000 c.p.s. as a 0 db reference level, the frequency response with a constant voltage input shall be within ± 2 db from 200 to 6K c.p.s. The response at 100 to 199 c.p.s. shall be within +2 to -3 db.

3.2.8.4.1.2.7 Internal volume control. - A volume control shall be provided in the amplifier module. With a 12.25 v. 1000 c.p.s. input, the volume control shall be capable of varying the output from 0 to 20 watts.

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3.2.8.4.1.3 Preamplifier AM-2120A/WIC. - Terminal designations for the AM-2120A/WIC shall be as follows:

Terminal	A-B	150-ohm input
	C-D	5-ohm input
	E	NC
	F	Ground
	G	NC
	H	+28 v. d.c.
	J	NC
	K	AGC input. *If other method of AGC is employed, this terminal may be deleted.*
	L	Amplifier output

The preamplifier shall be designed to drive the AM-2119A/WIC. The AM2120A/WIC and the AM-2119A/WIC, when electrically connected as shown on figure 4, shall meet all of the following requirements:

3.2.8.4.1.3.1 Input impedance. - The input impedance of the 150-ohm input shall be designed to accept the output of an M-99/U dynamic microphone element, or equivalent. The input impedance of the 5-ohm input shall be designed to accept the output of an M-87/U dynamic microphone or equivalent, or a low-power loudspeaker see 3.2.10 as a microphone. A frequency discriminating network as shown in control circuit schematic wiring plans for amplifier controls will be required in the 5-ohm input when the loudspeaker is used as a microphone.

3.2.8.4.1.3.2 Output impedance. - The output impedance requirements shall be the same as in the 1-watt module.

3.2.8.4.1.3.3 Direct current power input. - The combined amplifiers shall draw no more than 0.170 amperes at 28 v. d.c., when they are producing 1-watt signal output at 1000 c.p.s. at 25°C.

3.2.8.4.1.3.4 Gain. - The gain of the combined amplifiers shall be such that when the amplifiers are producing 1-watt output at 1000 c.p.s., the signal input to the 150-ohm input pad shall be between 0.36 and 0.81 volt, and the input to the 5-ohm input pad shall be between 0.14 and 0.28 volt (see figures 3 and 4).

3.2.8.4.1.3.5 Automatic gain control. - Automatic gain control shall be provided as follows:

(a) Break-point. - The point at which 1.0 db variation in input-output linearity occurs is defined as the break-point. The break-point shall occur between 1-watt and 2-watt output.
(b) AGC slope. - An increase in signal input of 20 db from the input level at the AGC break-point shall cause the signal output level to increase no more than 3.5 db from the signal output level at the break-point.
(c) Attack time. - The attack time shall be no greater than 0.2 second.
(d) Release time. - The release time shall be 7 ± 2 seconds.

3.2.8.4.1.3.6 Distortion. - The combined amplifiers shall meet the same distortion requirements as the 1-watt module.

3.2.8.4.1.3.7 Frequency response. - The combined amplifiers shall meet the same frequency response requirements as the 1-watt module, using input pad as shown on figure 3.

3.2.8.4.1.3.8 Input impedance. - The input impedances shall be of such a value that the combined amplifiers shall meet all of the requirements of this specification when a source impedance of 5 ohms resistive or 150 ohms resistive are applied to the 5-ohm or 150-ohm inputs respectively.

3.2.8.4.1.4 Cross talk due to power line modulation. - With a 0.2 volt r.m.s. signal between 200 and 6000 cycles per second superimposed on the 28 volt d.c. power supply terminals of the amplifier(s), the measured output at the amplifier audio output terminals shall be at least 40 db below rated power.

3.2.9 Signal alarm generators. - The signal generator modules shall have the following characteristics:

(a) Collision alarm: A simulated siren 750-1750 c.p.s. at three seconds per cycle.
(b) Missile alarm: A jump-tone signal alternating between 600 ± 10 c.p.s. and 1500 ± 15 c.p.s. at a rate of 6 c.p.s.

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(c) Jettison alarm: A recurrent sweep signal; each complete cycle starting at approximately 340 c.p.s. and rising continuously to an upper frequency of 2250± 100 c.p.s. The repetition rates shall be 100± 10 c.p.m.
(d) General alarm: A simulated gong striking at a rate of 100 strikes per minute.
(e) Nuclear plant casualty alarm: A jump tone signal alternating between 600± c.p.s. and 1500± 15 c.p.s. at a rate of 100 c.p.m.
(f) Diving alarm: A raucous signal similar to a klaxon horn consisting of approximately 445 c.p.s and 760 c.p.s.

3.2.9.1 Signal approval. - Alarm signals shall be recorded for a period of approximately 1 minute on a 7-1/2 inch per second 1/4 inch tape and submitted to the bureau or agency concerned for approval.

3.2.9.2 Alarm signal generator control circuitry. - Control circuitry shall be identical to figure 1. The plus 10-volt circuit and timer circuit are not required unless required by manufacturer's design. Interconnecting terminal board and connector plugs and jacks shall be marked to correspond and mechanically mate with required connectors in other parts of the electrical equipment cabinet.

3.2.9.3 Jettison alarm and general alarm signal duration. - Momentary closure of the alarm switches for these modules shall cause the signals to be generated for an adjustable period of either 10 or 15 seconds. Continuous closure of the general alarm switch shall have no effect other than to cause the alarm to generate continuously until the alarm switch is released. Operation of all other alarms specified herein shall require continuous closure of the alarm switches. Continuous closure of jettison alarm switch shall cause alarm to cease after 10 or 15 second duration. This shall release system for lower priority use.

3.2.9.4 Alarm signal adjustments. - Tone control adjustments shall be provided in an accessible location for each alarm module.

3.2.9.5 Signal generator module output. - Each signal generator module audio output shall be capable of driving the specified 1-watt amplifier AM2119A/WIC to rated output.

3.2.9.6 Complete test information including oscilloscope presentations at key test points shall be included in design drawings and technical manuals for each alarm signal.

3.2.10 Loudspeakers. - The following types of loudspeakers shall be used in equipment specified herein:

(a) Low power loudspeaker - LS-445 in accordance with Specification MIL-L-12606.
(b) High power loudspeaker - LS-211 in accordance with Specification MIL-L-25547.

3.2.10.1 Government furnished loudspeakers. - Loudspeakers used in the escape hatch announcing circuit (31MC) will be type LS-450/B. Loudspeakers to be supplied in portable bridge suitcase unit on the open bridge shall be type LS-445. These loudspeakers will be supplied by the Government to the shipbuilders, and are not used in any equipment covered by this specification, although they are to operate satisfactorily with the equipment specified herein.

3.2.11 Microphones. - Microphones, headsets, and handsets used in conjunction with the equipment specified herein shall be of the following types:

(a) IC/MPM-6 Hand held microphone in accordance with Specification MIL-A-20222. This microphone is supplied by the Government and is not to be furnished as part of the equipment required herein.
(b) Microphone M-99/U. - This is a noise cancelling microphone motor and is used in microphone assembly M-98/U, M-97/U, M-106/U, and M-114/U. Headset-microphone H-178/U, Headset microphone H-179/U, Handset H-177/U. Headset receiver shall be type H-157/U in accordance with Specification MIL-H-26312 modified to use M-99/U microphone element in lieu of M-87/AIC microphone element. Handset receiver element shall be type H-143/AIC.

3.2.11.1 Microphone M-99/U, when mounted in handset or headset assemblies of types specified herein, shall meet the following requirements:

3.2.11.1.1 Impedance at 1000 c.p.s. - Impedance at 1000 c.p.s. shall be 150 ± 15 ohms.

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3.2.11.1.2 Frequency response. - The microphone output shall fall within the limits shown on figure 5 when a sound pressure level of 20 microbars is applied at 1/4 inch from the microphone. There shall be no more than one peak or hole per octave.

3.2.11.1.3 Sensitivity. - Sensitivity at 1000 c.p.s. with sound pressure input of 100 microbars at a distance of 1/4 inch from artificial voice and output measured across 150-ohm resistive load shall be not less than 4 millivolts.

3.2.11.1.4 Microphone endurance. - The microphone shall be capable of withstanding 100 uninterrupted hours of operation with a voltage of 0.040 volts (r.m.s.) connected to the voice coil circuit. There shall be no lowering of sensitivity or other failure to meet the requirements of this specification when tested as specified in 4.5.9.4.

3.2.11.1.5 Noise cancelling requirements. - The random noise discrimination between 200 and 400 cycles shall be a minimum of 15 db.

3.2.12 AN/WIC system components and the overall system operational features. - Each of the system units shall function in the overall system as specified herein. Figure 1 consists of a set of 10 sheets of an overall system interconnect schematic diagram representing a typical POLARIS submarine installation of the SSB(N) type. For the list of stations, including equipment components and communication circuits required at each station to be incorporated in the development of the schematic diagram, see 6.1.

3.2.12.1 Intercom Set Control C-3047/WIC. - This is an amplifier control (talk-back control box) and is the basic station in the system. The amplifier control comprises the line (or distribution) amplifier, the loudspeaker for talking and listening, a spring return talk switch, four relays, a volume control for adjusting the listening level, an audio connector to permit the attachment of a headset-microphone accessory, and an indicator light which signals when the station or circuit is in use.

3.2.12.1.1 Detailed function of amplifier control in talk condition. - Refer to figure 1, sheet no. 9, (NAV. CENTER). The audio common circuit is connected to terminal 63/73. Terminal 70/80 connects to +28 volts and terminal 67/77 connects to -28 volts (ground). Operation of the press-to-talk switch S2101 to the talk position energizes relays K2101 and K2102 by connecting them to terminal 26 which is grounded by system, except during a higher priority operation. The loudspeaker circuit is disconnected from the line amplifier output and transferred to the five ohm input circuit through the equalization capacitor C2101 and the five ohm load composed of resistors R2108 and R2109 in parallel. K2101 transfers the 1-watt amplifier input from the isolation transformer T301 output to preamplifier output. Relay K2101 also connects ground to output terminal 65/75 which is the group control circuit. K2102 transfers the eight ohm output circuit of the line amplifier from the loudspeaker circuit to the preamplifier AGC circuit. This latter connection provides the AGC voltage for controlling the preamplifier so that changes in the input level due to different operators or the movement of the operator with reference to the loudspeaker will be minimized in the output. Relay K2102 transfers the 150 ohm output circuit of the line amplifier to the audio output circuit; terminal 62/72. Therefore, in the talk condition, the loudspeaker is connected as a microphone and the line amplifier is connected to the audio output line, and the group control circuit is energized. The group control circuit, by operating a master power relay at the distribution panel, applies amplifier power to terminal 69/79 which energizes the line amplifier. The amplifier, now being energized, permits the audio from the loudspeaker to be amplified to the 1-watt level for transmission as the group audio signal.

3.2.12.1.2 Detailed function of amplifier control in listen condition. - With the press-to-talk switch in the normal or listen condition, the group amplifier power circuit, when energized by another station, will energize the station line amplifier. The incoming audio, appearing on terminal 61/71, is connected to the station volume control R2101 through an isolation resistor R2104, R2101 in turn is connected to the internal level-set potentiometer R2103 and then to the input winding of the isolation transformer contained in the line amplifier. The output of the isolation transformer is connected through the de-energized contact of relay K2101 to the line amplifier input and after implication, the output appears at the 8-ohm and 150-ohm output terminals. The eight ohm output terminal is connected through the normally de-energized contact of K2102, through the linkage of terminals 35 and 25 and through the normal contacts of the press-to-talk switch S2101 to the loudspeaker voice coil. The 150-ohm output is connected through the normally de-energized contacts of K2102 to the primary winding of headphone transformer T2101. When no station on the circuit is talking, the transistor amplifiers are de-energized and draw no current from the power supply.

3.2.12.1.3 Detailed function of amplifier control using accessory equipment. - The amplifier control is arranged so that headphone and microphone accessories may be added. Terminals 6/16 and 7/17 provide

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for the connection of a low impedance, moving coil, microphone such as the M-34/AIC. The low side of the microphone circuit is connected to ground and directly to the low side of the 5-ohm input of the microphone preamplifier. The high side of the 5-ohm microphone circuit is connected to a switch section on S2101 and to the five ohm input circuit of the amplifier so that when the talk switch is in the normal position, the external microphone is connected through. If the S2101 is depressed to the talk position, the microphone will be disconnected from the five ohm circuit and the loudspeaker voice coil will be connected instead. The press-to-talk switch on the microphone will be connected from terminals 28 and 26 and operation of the press-to-talk switch will energize relays K2101 and K2102. Similarly, a 150-ohm microphone such as the Navy type IC/MPM-6 or the M-99/U series may be connected between terminals 3/13 and 4/14. The high side of this pair (terminal 4/14) connects through the talk switch S2101 and through the normally de-energized contacts of K2103 to the 150-ohm input of the microphone preamplifier. The microphone press-to-talk switch is connected between terminals 28 and 26 to energize the talk transfer relays and the control circuits. Microphone terminals 1/11 and 2/12 and relay K2103 provide for an additional microphone where it is necessary to disconnect the audio circuit or control another circuit independently of the normal control provided by the amplifier control. To obtain this operation, the additional circuit of the microphone press-to-talk switch is connected between terminals 26 and 38 and when operated, energizes relay K2103. Relay K2103 transfers the 150-ohm preamplifier input from microphone terminal 4/14 to microphone terminal 1/11. K2103 also closes through the control circuit between terminal 30 and terminal 40. These two terminals, which are independent of other circuits in the amplifier control, may be used for the control of an external circuit. A headset or a handset may also be connected. Either a 5- ohm or a 150-ohm microphone may be used when connected to the proper terminals as described for the microphone. The earphone or headset unit impedance should be in the order of 8 to 20 ohms and is connected between terminals 26 and 8/18. It will be noted that terminal 26 is also the low side of the control circuit. Terminal 8/18 is connected through the internal level adjusting resistance R2102 to the secondary of the headphone transformer T2101. When so connected, all incoming calls heard on the loudspeaker will also be heard in the earphone or the handset or headset. When talking out with the handset or headset, the earphone transformer remains connected to the 150-ohm output of the line amplifier. Operation transfer relay K2102 inserts a 1500-ohm resistance in series with the primary of this transformer, and thus sidetone will be heard in the headset or handset at a slightly reduced level. Headset jack J2103 is provided on the front panel of the amplifier control for the attachment of a headset. The microphone, receiver, and talk control circuits on this jack are in parallel with case terminal and operation, with respect to the microphone, the receiver, and sidetone, is as described in the preceding paragraph. When the headset extension cord is plugged into J2103, an internal jumper in the extension cable plug completes a muting connection so that if the amplifier control talk switch is placed in the extreme upper position, the loudspeaker of the amplifier control will be muted or silenced and incoming communications to the station will be heard only on the earphones of the headset. If the headset cable is unplugged without returning the press-to-talk switch to its normal position, the loudspeaker will be restored to service at a slightly reduced level. This is accomplished as follows. When S2101 is placed in the loudspeaker muting position, the high side of the loudspeaker voice coil circuit is connected by S2101 to terminal 33, through a linkage to 23 which is connected to terminal F of J2103, then through the plug jumper previously mentioned to terminal H of J2103 and to terminal 31 which is connected by linkage to ground terminal 21. Since the low side of the loudspeaker is also connected to ground, the muting circuit just described places a short circuit across the loudspeaker and prevents its operation. R2106 and R2107 in parallel provide a resistive load for the line amplifier when the talk switch is in the muting position. If the headset plug is removed, the common point connection between the loudspeaker and the resistive load is no longer grounded, and the resistor and the loudspeaker voice coil become connected in series to the 8-ohm output circuit of the line amplifier. By this means, audio output from the line amplifier will be heard in the loudspeaker even though the talk switch has been left in the muting position.

3.2.12.2 Intercom Set Control C-3048/WIC. - This is an amplifier control and the functions are identical to the amplifier control specified in 3.2.12.1, except that in addition, a momentary "Conn Report" switch is to be provided.

3.2.12.2.1 Detailed function of amplifier control in CONN report condition. - Refer to figure 1, sheet no. 5, (sonar supervisor). With the unit connected as specified in 3.2.12.1.1, operation of the CONN Report switch S2102 energizes relays K2101 and K2102 by connecting them to terminal 26. K2101 and K2102 transfer the unit from a listen to a talk condition as specified in 3.2.12.1.1. Input to a microphone, connected to terminals 3/13 and 4/14, will be amplified and transmitted as the group audio signal. In addition, S2102 closes the CONN Report control circuit connected to terminal 48 to group LO terminal 66/76, causing relays in the control cabinet and periscope station to operate. If the press-to-talk switch S2101 and the CONN Report switch S2102 are operated simultaneously, the loudspeaker will be connected as a microphone (as specified in 3.2.12.1.1) and transmit the audio to the circuit set-up by the CONN Report circuit.

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3.2.12.3 Intercom Set Control C-3323/WIC. - This is an amplifier control, and the functions are identical to the amplifier control specified in 3.2.12.1, except that in addition, a two-circuit selector switch is to be provided for selection of either or both of the two circuits.

3.2.12.3.1 Detailed function of amplifier control. - Refer to figure 1, sheet no. 10, (preparation panel). The operation of this unit is the same as that specified in 3.2.12.1.1 and 3.2.12.1.2, except that switch S2102 is provided to control communication on two sub-circuits. With the switch in the center (A and B) position, the audio output of the line amplifier is connected through S2102-7 and 11 to the group A audio line at terminal 61/71, and through S2102-8 to the group B audio line at terminal 45. The audio input circuit is connected to the audio output circuit by a link between terminals 41 and 51. The control circuits are connected through S2102-6 and 10 to the group A control circuit at terminal 65/75, and through S2102-5 and 9 to the group B control circuit at terminal 46. Each audio line is loaded with a 330-ohm resistor. When switch S2102 is placed in the upper (Group A) position, the audio and control circuits of group B are disconnected, and the station is connected to group A only. With switch S2102 in the lower (Group B) position, group A circuits are disconnected.

3.2.12.4 Intercom Set Control C-3224/WIC. - This is an amplifier control, and the functions are identical to the amplifier control specified in 3.2.12.1, except that in addition, a two-circuit selector switch shall be provided for selection of either of the two circuits.

3.2.12.4.1 Detailed function of amplifier control with circuit selector switch. - Refer to figure 1, sheet no. 7, (fire control coordinator). The operation of this unit is the same as that specified in 3.2.12.1.1 and 3.2.12.1.2, except that switch S2102 is provided to select either of two circuits. With the switch in the upper (27MC) position the audio output of the line amplifier is connected through S2102-2 and 5 to the 27 MC audio line at terminal 61/71. The audio input circuit is connected to the audio output circuit by a link between terminals 41 and 51. The control circuits are connected through S2102-3 and 6 to the 27 MC group control circuit terminal 65/75. In addition, S2102 closes the Torpedo Room audio circuit terminal 45 to terminal 39 which may be connected to a monitor loudspeaker. With switch S2102 in the lower (47MC) position the audio lines are reversed and the control circuit is connected through S2102-6 and 9 to the 47MC circuit at terminal 46.

3.2.12.5 Intercom Set Control C-3051/WIC. - This is a 9-station control box and does not in itself provide any audio amplification, but shall be used in conjunction with an amplifier control for talking and listening on a selective circuit. In some cases, a monitor loudspeaker will be necessary to provide monitoring of one circuit while the station is being used for communication on another circuit. The 9-station control box is to contain 10 push-button selector switches providing for nine active selections and one stand-by position. The push-button switches are mechanically interlocked so that no two push-buttons can be operated simultaneously. Five toggle switches shall be provided which control various monitoring functions, a separate volume control for adjusting the level of the monitoring circuit. A red illuminated edge lighted panel shall be provided so that the panel markings are more readily visible when the unit is used in darkened areas. The front panel dimmer control is to determine the amount of illumination.

3.2.12.5.1 Detailed function of 9-station control box. - Refer to figure 1, sheet no. 8, (radio room position no. 1). The relay coil circuit is wired through a series arrangement of monitor switches S1704 and S1705 to the station intercom call-in circuit, terminal 54, and to a buss on the selector switches. A control voltage appearing on the call-in circuit, terminal 54, will energize relay K1701. In the energized position, K1701 applies +28 volts to terminal 65/75. Terminal 65/75 is externally connected to the amplifier power terminal of the amplifier control and will energize the station amplifier. In the de-energized position, K1701 connects terminal 64/74 to amplifier power terminal 65/75 so that the application of +28 volts to the group power lines at the central control cabinet will energize the station amplifier power circuit. Operation of either monitor switch S1704 or S1705 will energize relay K1701 and apply power to the station amplifier. Relay K1702 is the circuit control relay. The relay coil control circuit is connected to terminal 61/71, which is connected externally to the talk control circuit of the amplifier control. Relay K1702 will be energized each time the station press-to-talk switch is operated. Thus operated, K1702 provides a means for energizing the control circuit of the selected communication circuit. K1702 also energizes a muting circuit which may be used to control a relay in an adjacent station or loudspeaker line on the same circuit, to prevent acoustical feedback. Relay K1703 normally completes the intercom audio call-in circuit, terminal 53, and the audio in and out circuit, terminal 1/11. Terminal 1/11 is connected to the corresponding terminal in the amplifier control. When the jumpers on E1703 are connected so that the first two selector switch positions are arranged for communication on the talk-back loop circuit, the jumper connections noted will cause relay K1703 to be energized whenever either of the circuits are selected. Operation of relay K1703

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transfers the intercom call-in audio circuit to the station monitoring system through resistor R1709 and monitor volume control R1701. Thus, if the station has selected a loop circuit, an incoming intercom call will be received and the call will be isolated from the selected communication circuit. If the jumpers on E1703 are arranged so that no talk-back loop circuit connection is provided, the jumpers which energize K1703 are not connected and the relay only completes the normal audio circuit. When an intercom station is selected, the selected station's audio circuit is connected to an audio buss which terminates at terminal 1/11. The selected station control circuit is connected through a pair of contacts to two control busses. The power buss terminates at 62/72, the negative side of the power supply. The other buss terminates 54, the local station intercom call-in control circuit. The control lead of the selected station is connected externally to the intercom call-in terminal of the desired station. The control thus completed, energizes relay K1701 and the corresponding power relay in the called station, applying power to the amplifiers of both stations. Operation at either station will energize the audio frequency circuit and the communication will be heard at the listening station. When the transmission is completed, the calling station releases the selection by operation of the standby selector switch which de-energizes the control buss, removing the amplifier power from both stations. On an incoming call, the originating station has energized both stations and the called station need only to select stand-by to reply. Operation of the talk/listen switch is then all that is required. If the jumpers on E1703 are connected for talk-back on loop circuits, operation of the chosen selector switch will connect the audio line to the audio buss as described for intercom operation. The selected circuit control line will be connected to a contact on relay K1702. The operating lead of K1703 will be connected to the power control buss, causing the relay to remain operated as long as the push-button switch is depressed. The operation of K1702, when a press-to-talk switch is operated, will now energize the group control circuit. This group control circuit energizes the busy signals at all the talk-back stations connected to the circuit and energizes the group power relay in the control cabinet. The group power relay, in addition to energizing the amplifiers of all the stations on the circuit, also feeds the amplifier of the originating station through the normally closed contacts of K1701. At each selector switch, a normally closed contact is provided in connection with the audio circuit so that by connection through a monitoring switch and resistor as shown with S1701 and R1704, the circuit may be connected to the station monitoring system. The number one push-button is connected directly through monitoring resistor R1703, which provides for the monitoring of the circuit connected to that button regardless of other uses of the station. The circuit connection thus normally provided, allows for the connection of a prime circuit to the number one push-button which will be continuously monitoried, and the connection of a secondary circuit to the number two button which may be monitored at will. The No. 9 push-button monitoring contact is connected to R1713 and provides for continuous monitoring of this circuit. R1713 is removed at all stations except the Ballast Control Panel, where continuous monitoring of the Deck Watch station is required. At stations where no talk-back loop circuits are connected, R1703 is removed to eliminate the continuous monitoring of the circuit on No. 1 push-button. Monitoring switch S1203 provides a special facility for the control of the power to a monitoring loudspeaker which may derive its audio signal from a source independent of the communication station. This connection is used in the Radio Room where the audio input to the radio monitor loudspeaker is obtained from the radio receiving circuits. Monitor switches S1704 and S1705 provide facilities for the monitoring of communication circuits which do not energize central power circuits. An example of this is when S1705 is used for monitoring the radio receiver audio distributed by the AN/WIC system. The audio input is connected to terminal 5/15 which is connected through S1705 to monitoring isolation resistor R1708 and then to monitor volume control R1701. As noted previously, the operation of S1705 will energize the station power circuit. Monitor volume control R1701 is connected through isolation resistor R1711 to terminal 4/14. Terminal 4/14 is connected either to a separate monitor loudspeaker or to the monitoring input terminal 59 of the amplifier control. R1710 serves as a load resistor on the station audio circuit. When two stations have been connected for inter-communication, the load resistor in each station is paralleled and a load of approximately 150 ohm is applied to the audio circuit. The circuit connection on stand-by switch S1706K functions only in connection with intercom operation to the Bridge station. As was specified, when interior stations operate on the intercom circuit amplifier, power is continuously applied to both stations and the called station may reply without making a selection. However, the Bridge station amplifier power is only available when the Bridge press-to-talk switch is operated. When the Bridge station originates a call on the intercom circuit, in order that the called station may reply without making a selection, a means shall be provided for the called station to energize its own and the Bridge amplifier power when it replies. The circuit connection on the stand-by switch position provides this by making a connection between the intercom call-in control buss and the control circuit contact of K1702. Since relay K1702 operates each time a talk switch is depressed, the intercom call-in control line is grounded energizing K1701 and causes it to energize the local station amplifier. The intercom call-in control line energizes a corresponding relay in the Bridge control box, which in turn energizes the Bridge station amplifier. If two intercom stations are adjacent and experience acoustical feedback when both stations have selected the same circuit, the corresponding "P" circuit at each station should be connected together. The loudspeaker muting circuit, terminal 56, should be connected to the relay coil,

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terminal 42, of the amplifier control at each station. One set of the auxiliary relay contacts at the amplifier control will be connected so as to mute the loudspeaker. With this circuit arrangement, whenever the two stations which are interconnected select the same circuit, a connection is completed between the muting relay at each station and the normally open pair of contacts on the switch box talk control relay (K1702). When a talk switch at either station is operated for a transmission, both muting relays will be energized. The muting relay at the transmitting station will be ineffective since the loudspeaker has already been disconnected from its output circuit for use as a microphone. The muting relay at the other station will operate to silence its loudspeaker.

3.2.12.6 Intercom Set Control C-3054/WIC. - This is a 15-station control box and the functions are similar to a 9-station control box specified in 3.2.12.5, except for the larger number of selections.

3.2.12.6.1 Detailed function of 15-station control box. - Refer to figure 1, Sheet No. 9, (Navigation Center). The 15-station control box functions basically the same as the 9-station control box specified in 3.2.12.5.1. This box, however, has not been arranged for conversion to different modes of operation by changing internal wiring jumpers. The box is wired so that the first four switch positions (A through D) will be used on talk-back loop circuits requiring the control of external circuits. No. 8 position (H), is wired for use with the 31 MC system. The balance of the switch positions (5 through 7 and 9 through 15) are connected for use in the intercom system. In this unit, relay K2601 is the station power control relay and functions as described for relay K1701 in the 9-station unit. The corresponding call-in control terminal is 56 and the corresponding group power terminal and amplifier power terminal are 68/78 and 69/79, respectively. Relay K2603 is the circuit control relay and functions the same as its counterpart K1702 previously described. Relay K2602 is the audio transfer relay and performs the same functions as specified for relay K1703. Relay K2604 has no counterpart in the 9-station control box. It provides a means for the independent control of a special circuit, and when energized, will break the control circuit and transfer the audio circuit from the selector switch busses to a special output terminal. The 15-station control box is provided with automatic monitoring of the circuits connected to button No. 1 and button No. 4. Selective monitoring is provided for buttons No. 2, No. 3, and No. 8 by S2601, S2602, and S2603. Operation of any monitor switch, S2603, S2604, or S2605, will energize relay K2601 and apply power to the station amplifiers. It should be noted that at any position where monitoring, either automatic or selective, is not desired, the monitoring may be eliminated by removing the isolation resistor applicable to the circuit.

3.2.12.7 Intercom Set Control C-3502/WIC. - This is a 15-station control box and the functions are nearly identical with that of the 15-station control box sepecified in 3.2.12.6, except for the addition of a radio handset receptacle J2803 and a carrier-on indicator light DS 2805 with a series dropping resistor R2814. As shown on figure 1, sheet No. 9, (Missile Control Center), these items are independently connected to a series of terminals from whence they may be connected to a radio telephone remote control box or to the Radio Room switchboards, and to an amplifier-loudspeaker unit for the monitoring of the radio receiver circuit. The amolifier loudspeaker is identical with those used in the AN/WIC Intercommunication System and contains its own integral transistorized audio amplifier. The 28 volt power for operating the loudspeaker amplifier is obtained from the AN/WIC system by means of the on-off control switch S2803.

3.2.12.8 Intercom Set Control C-3053/WIC. - This is a 15-station control box used for missile launcher operations. The functions are similar to that of the 15-station control box specified in 3.2.12.6. The major difference is that push-button switches S2706A through S2706C and S2706J through S2706M, refer to figure 1, sheet No. 10, (Launcher Operations station), are wired for use on talk back loop circuits requiring the control of external circuits. The latter four switches are used for control of the 35 MC circuit in the missile area intraspace communication network. The muting (P) circuits of S2706J and K are combined so that either switch will energize the group 1 muting circuit at terminal 2/12; likewise either S2706L or M will energize the group 2 muting circuit at terminal 4/14. The sub-group audio circuits are connected through normally closed contacts of the selector switch (S2706) to monitor switches S2701 through S2704. The remaining monitor switch, S2705, is wired for connection to the AN/WIC radio monitoring system through terminal 10/20. The power control section of the switch is arranged to operate relay K2701 in order to apply power to the station amplifier.

3.2.12.9 Intercom Set Control C-3050/WIC. - This is a two circuit control box used for control of the 31 MC circuit escape trunk announcing system. This control box contains a 20-watt amplifier dor driving the escape trunk loudspeaker, a line amplifier used when the transmission is being made from the escape trunk, a local station circuit selector switch, and two relays for circuit control and transfer.

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3.2.12.9.1 Detailed function of 31 MC control box. - Refer to figure 1, sheet No. 7, (FWD. ESC. TRUNK) 31 MC control box. The storage battery is connected to terminals 21/31 nnd 22/32. The positive terminal (22/32) is connected through CR2001 to the system -28 volt power on terminal 42/52 when ESCAPE TRUNK switch S2002 is in either the OFF, or ON position. In the EMERG position the battery is disconnected from the system power circuit and supplies emergency power. Switch S2002 also disconnects external 31 MC audio and control lines in the EMERG position. The escape trunk amplifiers and control relays are connected to 28V power in the ON and EMERG positions of S2002. LOCAL STA switch S2001 is a two-position selector for the associated station. When the switch is in the 7 MC position, the amplifier control unit is connected to the normal 7 MC circuit and functions as a standard talk-back station on this circuit. With S2001 in the 31 MC position, -28 volt power on terminal 70/80 is connected directly to the amplifier-terminal 69/79 to energize the line amplifier of the amplifier control unit. Amplifier control group LO, connected to terminal 66/76, is disconnected from the 7 MC LO and connected directly to ground terminal 67/77, thus removing the 7 MC priority control from the unit. The talk control connection on terminal 65/75 is transferred from the 7 MC control circuit terminal 48 to the relay coil circuit of K2001 and K2002, and through S2002 in the OFF and ON positions to the 31 MC control circuit on terminal 47. The audio circuit from the amplifier control, connected to terminal 61/71, is disconnected from the 7 MC audio terminal 50 and connected through S2002 to the 31 MC audio terminal 49. This circuit also connects through the normally closed contact of K2001 to the 150 ohm output of line amplifier A2001. With switch S2001 in the 31 MC position, 7 MC audio is connected to the monitor terminal 59 through the isolation resistance R2002, so that a call on the 7 MC circuit may be received. With switch S2002 ON, +28 volt power is applied to the relay coils and to the arm of K2001. In the de-energized position, K2001 applies power to line amplifier A2001. The Escape Trunk loudspeaker is connected to terminals 1/11 and 2/12 through the contacts of K2002 to the line amplifier input, loaded by resistors R2004 and R2005 in parallel. In this condition, the Escape Trunk line amplifier and the associated amplifier control unit are energized and speech into the loudspeaker in the Escape Trunk will be connected to the 31 MC audio line and will be heard on the amplifier control unit. When the amplifier control unit replies by operating a press-to-talk switch, the 31 MC control circuit will energize relays K2001 and K2002. The Escape Trunk loudspeaker is transferred by relay K2002 to the 16 ohm output of the 20-watt audio amplifier, A2002. Relay K2001 transfers +28 volt power from the line amplifier to the loudspeaker power amplifier and disconnects the 150 ohm output of the line amplifier from the audio circuit. The escape trunk loudspeaker will receive any communication from the local amplifier control which may appear on the 31 MC audio circuit. The audio signal is applied to the input of power amplifier A2002 through isolation resistance R2001 and volume control R2003. A one ohm resistor (R2006) in series with the volume control prevents the reduction of the incoming volume to zero. Resistor R2007 provides a partial load on the 31 MC circuit. The local station and escape trunk amplifiers are protected by fuse F2001. Diode CR2001 permits the system to supply power and charge the auxiliary storage battery as required, but prevents the auxiliary battery from discharging into the system power lines if the main power source should fail. If the 31 MC audio terminal (49) and the 31 MC control terminal (47) are connected through the system to the control room stations, selection of the 31 MC circuit at the periscope station and the diving officer station will permit communication between the control room, the escape trunk, and the escape trunk control station in the adjacent compartment. Communication between the control room and the escape trunk will be independent of the position of switch S2001 on the 31 MC control box. Test jacks J2004 and J2005 have been provided on the front panel of the 31 MC control box to provide for checking the voltage of the auxiliary battery and the line. With switch S2002 in the ON position, the jack measures the line voltage of the system power source. With the switch in the emergency position, the tip jacks measure the battery voltage only. When the escape trunk is not in use, the LOCAL STA switch S2001 should be in 7 MC position, and the ESC TRUNK switch S2002 should be left in the OFF position.

3.2.12.10 Intercom Set Control C-3049/WIC. - This is a periscope/bridge control box containing control circuits and selector relays for operation from the periscope area or from the portable ships control unit on the bridge or other topside circuits.

3.2.12.10.1 Detailed function of periscope/bridge control box. - Refer to figure 1, sheet No. 3, (Bridge and Periscope station). On the upper part of the front panel are the switches and volume controls for the Bridge station. Topside switch S1911, when in the off position, disconnects all active control and audio wires to the bridge and prevents the operation of the relay circuits which make connection to the bridge. Direct radiator loudspeakers and selector switches required for communication at the bridge are contained in the portable ships control unit. One of these switches is for the 1 MC and 7 MC circuits only. The other switches are a multiposition selector switch and a press-to-talk switch which provide for a selection of tactical or intercommunication circuits. The position of the selector switch has no effect on the operation of the 1 MC or 7 MC circuits.

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3.2.12.10.2 Bridge - 1 MC. - The 1 MC circuit on the bridge is the highest priority voice circuit. Closure of the 1 MC switch energizes relay K1913. Relay K1913 energizes relay K1917, relay K1910 and Bridge Busy Light DS1901. Relay K1913 breaks the supply circuit to terminal 10/20 which prevents periscope station 1 MC microphone operation. Relay K1913, through normally closed contacts of relay K1903, also energizes relays K1901 and K1911. K1917 energizes the 1 MC control circuit in the central control cabinet. K1917 also operates K1923 and K1938. K1923 prevents the operation of the periscope station. K1938 prevents bridge selector switch talk operation and is connected to mute the input to the bridge monitor amplifier No. 2 in order to prevent interference with the 1 MC transmission. A "conn report" or "torpedo detection" message will be received on the bridge monitor speaker. K1917 connects the 150-ohm output of the bridge line amplifier to 1 MC audio output terminal 149/159, where it is connected through ship's wiring to the control cabinet for distribution through the system. K1910 disconnects the power from the 20-watt amplifier No. 1. K1901 connects the bridge talk/listen loudspeaker through capacitor C1901 to the contacts of relay K1911. K1911 closes this circuit through to the 5-ohm input terminals of the Bridge line amplifier. The energizing of the 1 MC control and master circuits, having operated the system power relays in the distribution panel of the control cabinet, causes +28 volt power to appear at terminal 169/179. Terminal 169/179 is connected through the normally closed contacts of relay K1933 to the +28-volt connection of the bridge line amplifier. With this sequence of events, speech into the bridge talk-listen loudspeaker will be amplified by its line amplifier and distributed over the 1 MC audio circuit for broadcast by all connected loudspeakers. When 1 MC announcements are made from the interior of the ship, or when alarm signals are being transmitted over the 1 MC system, +28-volt power is again applied to terminal 169/179, thus energizing the bridge amplifiers. 1 MC audio signal is now present on terminal 149/159. This audio circuit is connected through isolation resistor R1935 to terminal 221/231 where it is connected by ship's wiring to the amplifier No. 1 input terminal 30/40 of the Bridge listen amplifier. The 8-ohm output of this amplifier is connected through a 16-ohm series resistor, R1601, to terminals 227/237 and 228/238 which are connected to relay K1910. With relays K1910 and K1901 in the de-energized position, the amplifier output circuit is connected through to the bridge talk-listen loudspeaker and the 1 MC audio signal will be reproduced by this loudspeaker. The 1 MC audio signal will be received at a fixed level regardless of the position of the bridge listen volume control R1901.

3.2.12.10.3 Bridge - 7 MC. - When the bridge 7 MC switch is closed, relays K1915 and K1916 are energized, through the de-energized contact of K1923. K1923 provides for priority of the 1 MC microphones over 7 MC operation from the bridge. K1915 energizes the bridge busy signal, relay K1910, and through the normally closed contacts of K1903, energizes K1901 and K1911. K1915 energizes the Conn 7 MC control circuit terminal 181/171 and opens the Periscope LO circuit terminal 186/196 preventing operation of the periscope station. K1916 completes the 150 ohm output circuit from the line amplifier to the 7 MC audio circuit on terminal 21. The path of this circuit is from the line amplifier through the de-energized contacts of K1906 to the contact of K1916. Relays K1901, K1910, and K1911 function as described for the 1 MC system to transfer the bridge talk/listen loudspeaker from the output of its power amplifier to the input of its line amplifier. The energizing of the 7 MC control circuit in the central control cabinet in turn energizes the 7 MC power distribution relays energizing all the station amplifiers connected to the 7 MC circuits. Speech into the bridge talk/listen loudspeaker will be amplified and distributed over the 7 MC audio circuit. Incoming 7 MC messages to the bridge are received on the monitor speaker as follows: Operation on the 7 MC circuit is made by one of the interior stations having energized the power distribution relays. Power will be applied to the bridge amplifiers in the same manner as for the 1 MC operation previously described. The 7 MC control connected to 162/172 causes relay K1937 to operate. The 7 MC audio communication now appearing on terminal 21 is applied through the de-energized contacts of K1916 through the circuit isolation resistor R1908, through the contacts of K1937 to the de-energized contacts of priority relay K1931 and bridge monitor volume control R1902. R1902 is connected to terminal 222/232 and through interconnecting wiring to terminal 50/60 of the amplifier No. 2 input of the bridge monitor loudspeaker amplifier. The eight-ohm output of amplifier No. 2 is connected through a 16-ohm series resistor, R1602, and the topside disconnect switch to the bridge monitor loudspeaker.

3.2.12.10.4 Bridge - selector operation. - By making selection on the bridge selector switch and using the associated press-to-talk switch, communication may be accomplished on the 27 MC circuit and to several interior intercommunication stations. With the bridge selector switch on 27 MC, relays K1918 and K1940 are energized. K1918 completes the connection between the 27 MC control circuit on terminal 32 and talk relay K1935. K1918 disconnects the 27 MC audio circuit on terminal 31 from its monitor isolation resistor, R1907. K1940 completes the connection between the 27 MC audio circuit on terminal 31 and the receiving isolation resistor R1910 connected to bridge listen volume control R1901. K1940 transfers the intercom call audio circuit on terminal 135 from isolation resistor R1910 to isolation resistor R1909 connected to the bridge monitor volume control through K1937 and K1931. This connection assures

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the receipt of intercom calls to the bridge when the 27 MC circuit has been selected and isolates the 27 MC circuit from the intercom circuits. Operation of the selector switch talk switch energizes K1907 and K1935. With the 27 MC circuit selected, K1907 completes the audio connection from the 150 ohm output of the bridge line amplifier through de-energized contacts of K1906 and energized contacts of K1940 to the 27 MC audio line (31). K1907 also prevents the bridge listen power amplifier from becoming energized when the system power is applied to the station. K1935 energizes K1910, K1901, K1911 and the bridge busy light. K1935 energizes K1933 and the circuit control buss of the bridge selector relays and thus through the energized contact of K1918, the 27 MC control circuit on terminal 32 becomes energized. The 27 MC control circuit in turn energizes the system power relays, applying power to the station amplifiers. K1936 energizes relay K1933. K1933 applies power to the bridge station amplifiers. Relays K1901, K1910, and K1911 function to transfer the bridge talk/listen loudspeaker into a microphone as previously specified. Speech into the loudspeaker will be amplified and transmitted to all connected 27 MC stations. Incoming 27 MC messages are received on the talk/listen loudspeaker, when the selector switch is allowed to remain in the 27 MC position. Operation by an interior 27 MC station will energize the system power relays causing amplifier power for the bridge loudspeaker, appearing on terminal 169/179, to be connected through the de-energized contacts of relay K1933 and the de-energized contact of K1907 to the bridge listen power amplifier. The 27 MC audio circuit connected through isolation resistor R1910 as previously noted, will be amplified by the power amplifier and transmitted to the bridge loudspeaker. If the 27 MC position has not been selected, messages on the 27 MC circuit will not be received on the bridge unless the bridge 27 MC monitoring switch S1903 or priority relay K1931 have been operated. K1931 is the Conn report priority relay and is energized whenever a 27 MC station selects the Conn reporting circuit. The operation of this circuit at a reporting station also energizes the system power relays in the same manner as for normal 27 MC circuit usage. Relay K1931 then functions to connect the 27 MC monitoring resistor R1907 to the monitor volume control R1902 from which the signal is fed to the monitor amplifier and thence to the monitor loudspeaker.

3.2.12.10.5 Bridge - selector intercom. - Selection of an intercom circuit at the selector switch will energize one of the selector relays. Each relay completes a particular intercom audio circuit from its terminal to the receiving isolation resistor R1910 and to the normally open contact of relay K1907. Each relay also completes the corresponding intercom control circuit from its terminal to the normally open contact of talk relay K1935. When the selector switch press-to-talk switch is operated, relays K1907 and K1935 are energized and perform the same functions as specified for the 27 MC circuit, except that now an intercom audio circuit becomes connected by the closure of K1907 and K1936, and an intercom control circuit becomes energized by the closure of K1935. Operation of K1936 will energize the bridge station amplifier by energizing power relay K1933 as previously specified. The selected control circuit energizes the power relay in the called station and applies power to the called station amplifier so that it may receive the communication from the bridge. When the called station replies, power will be applied to the Conn station call-in circuit on terminal 136, thereby energizing relay K1933 and applying amplifier power to the bridge listen amplifier. When the topside switch is in the "ON" position, the periscope station intercom receiving circuits are disconnected from the conning station intercom call-in circuit. Communication to and from the bridge will not be heard at the periscope station unless the periscope station selects the same station. With IN PORT selected, both Periscope and Bridge may receive and answer calls. Four of the five intercom circuits available at the Bridge are connected in parallel with the push-button selector in the periscope-bridge control box. The fifth circuit is connected to the COC intercom circuit by terminals 141/151 and 142/152. The remaining circuit is designated as a SEL circuit and is connected to terminals 137 and 138. This circuit maybe interconnected at the terminal board with the 32 MC announcing circuit or with any 21 MC intercommunication circuit. While the Bridge is transmitting over intercom circuits, the audio connection is also completed through the de-energized contacts of relay K1940 to the CONN intercom audio line so that any interior station selecting the Bridge while this transmission is going on, will be aware that the Bridge is busy and should not be interrupted. When the Bridge selector switch is in an intercom position, incoming messages are received on the Bridge talk/listen speaker in the same manner as was described for the 27 MC circuit. If the SEL circuit is used for 32 MC, terminal 139 is connected to terminal 108/118 so that relay K1939 will operate during 32 MC selection at the bridge. K1939 transfers the intercom call audio circuit on terminal 135 to isolation resistor R1909 connected to the Bridge monitor volume control through K1937 and K1931. If 32 MC is not selected, K1932 becomes de-energized and the 32 MC audio circuit is connected through the de-energized contact to isolation resistor R1926, to the monitor volume control, and the input of the monitor loudspeaker.

3.2.12.10.6 Bridge - microphone receptacle. - With a portable hand-held microphone (M-114/U) plugged into the receptacle on the portable ships control unit, communication will be possible on the 7 MC circuit. This is accomplished by a relay and transformer mounted in the bridge unit. When the microphone

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talk switch is operated, one set of contacts closes the control circuit to the relay in the bridge unit. This relay connects the microphone, through the matching transformer, to the T/L input circuit in lieu of the loudspeaker element. The other set of microphone contacts close the 7 MC control circuit which operates as specified in 3.2.12.10.3. The circuitry associated with the bridge microphone receptacle selector switch S1912 is now not utilized. The position of this switch is therefore immaterial.

3.2.12.10.7 Deck watch operation. - During deck watch operation the portable ships control unit may be moved from the bridge to the deck watch position. An extension cable, provided by the shipyard, will connect between the unit and the cable plug located in the bridge. For this mode of operation the topside switch should be in the ON position. If the topside switch is set to the IN PORT position 1 MC and 7 MC operation will be as specified in 3.2.12.10.2 and 3.2.12.10.3. The bridge talk-switch, however, will now operate relays K1904 and K1914. If 1 MC is not operated K1904 will operate the CONN I/C CALL IN, the power-relay K1933 and the bridge talk relays K1901, K1910 and K1911. Relay K1914 operates the BCP control line and relays K1908, K1909, and K1912. K1909 and K1912 serve no function. Relay K1908 connects the output of the line amplifier to the BCP audio line. Therefore operation of the talk switch will provide communication to the ballast control panel regardless of the setting of the selector switch on the bridge.

3.2.12.10.8 Periscope operation - bridge on. - When the ship is operating on the surface and being conned from the bridge, the topside control switch will be set at ON as was specified previously. The periscope station may be used to communicate with any of the circuits or intercom stations indicated on its selector switch. The 1 MC microphone at the periscope station is subject to the higher priority of the 1 MC voice from the bridge. There is no priority on 7 MC operation. When the bridge is off and the "conn" of the boat has been transferred to the control room, the periscope station has first voice priority on the 1 MC circuit, and also has priority of the 7 MC circuit with respect to the talk-back stations on this circuit.

3.2.12.10.9 Periscope operation - 1 MC. - Two microphones are provided in the control room for use exclusively on the 1 MC circuit. One of these microphones is located at the periscope station; the other microphone is located at the ballast control panel. The 1 MC microphones are connected to the periscope station control box 1 MC circuit through a microphone interconnecting box J-1072A/WIC. The interconnecting box is connected to terminal 7/17, 8/18, 9/19, 10/20 and 190/200. Terminal 190/200 supplies the +28 volts d. c. for operating the relay in the microphone interconnecting box. Terminal 10/20 is connected through the normally de-energized contacts of relay K1913 to the Conn LO circuit on terminal 166/176. As specified for bridge operation, this circuit is connected to ground through the emergency reports priority relay and the alarm signal generator priority relays in the control rack. K1913 can only operate to disconnect the perioscope microphone control circuit when the bridge is on. Closure of the periscope 1 MC microphone talk switch completes the circuit to terminal 24/34 and 25/35 of the microphone interconnecting box. This operates relay K2301 and completes the circuit through terminal 9/19 and to relays K1921 and K1922. K1921 energizes relay K1917 and prevents the operation of open bridge relays K1902 and K1903. Relay K1922 completes the circuit from the microphone input terminals 7/17 and 8/18 to the 150 ohm input of line amplifier, A1901. Relay K1917 energizes K1923, and interrupts the 1 MC audio circuit to terminal 184/194 and the periscope control circuit which, as previously specified, for station amplifier control to prevent acoustical feedback. Relay K1917 also connects the 150-ohm output of the line amplifier to the 1 MC audio circuit and energizes the 1-MC control circuit which, as previously specified for bridge operation, energizes the system power relays and establishes the necessary connections for 1 MC operation. If the topside is ON or if the deck watch station has been connected, the operation of K1923 will also interrupt the control circuits to the bridge to prevent bridge operation on either the 7 MC circuit or selector switch circuit, thereby establishing the priority of the periscope 1 MC operation over system voice operation of lesser priority. With the 1 MC audio and control circuits completed as specified, speech into the periscope 1 MC microphone will be amplified and transmitted to all 1 MC receivers. The ballast control panel 1 MC microphone is connected to terminals 4/14, 5/15, 25/35 and 23/36 of the 1 MC microphone interconnecting box. The microphone circuit is completed through the normally de-energized contacts of relay K2301 which provides priority for the periscope station microphone. The microphone control circuit is connected to periscope station terminal 9/19. Operation on the 1 MC circuit from the ballast control panel is as specified for the periscope station.

3.2.12.10.10 Periscope - 7 MC. - To operate on the 7 MC circuit, this circuit is selected on the push-button selector switch. Closure of this switch (S1910A) connects the 7 MC audio circuit on terminal 21 to the selector switch audio buss which is connected through terminals 181/191 and 182/192 to the periscope station amplifier control unit. The 7 MC control circuit on terminal 22, cross connected with terminal 161/171, is connected to the 7 MC control relay in the control rack. The closure of the 7 MC selector switch also energizes relay K1924. K1924 energizes relay K1926 and prevents the operation of relay K1934. K1934 establishes

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the priority of 7 MC circuit monitoring over other circuits and has no function when the 7 MC circuit is selected at the periscope station. K1926 transfers the Conn intercom audio call-in circuit from the selector switch audio buss to the monitor system through isolation resistor R1915. This transfer isolates the normal talking circuit from the intercom circuit and provides for the receipt of incoming intercom calls during operation on the 7 MC circuit. The selector switch control buss is actuated by the operation of relay K1925. K1925 is connected by way of terminal 185/195 to the talk control circuit in the amplifier control unit. Closure of a talk switch in the amplifier control unit will energize talk relay K2101, thus energizing relay K1925. The closing of K1925 energizes the 7 MC control system and turns on the amplifier power for the 7 MC stations. Speech into the amplifier control loudspeaker or into a microphone associated with the station will be transmitted to the selector switch audio buss and through the 7 MC selector switch to the audio circuit, for distribution to the 7 MC stations. A reply or communication on the 7 MC circuit is received at the periscope station when the calling station energizes the system. If the 7 MC selector switch is operated, audio will be connected to the amplifier control. If the 7 MC selector switch is not in the operated position, K1924 is not energized, and operation of the 7 MC control circuit energizes K1934. The 7 MC audio circuit is connected through the normally closed contact of S1910A to monitor isolation resistor R1916, where connection is made through K1934 and the Conn Report priority relay to monitor volume control R1903 and the monitor loudspeaker input circuit on terminal 81/91. Other monitor circuits are grounded during 7 MC operation. The periscope monitor loudspeaker amplifier is energized whenever the 7 MC circuit is operated and, therefore, the 7 MC transmission will be heard on this monitor loudspeaker. Operation on the 27 MC circuit using switch position S1910B is similar to that specified for the 7 MC circuit. Closure of this switch energizes relay K1926, transferring incoming intercom audio to the monitor loudspeaker and isolating the intercom circuit from the selected circuit. When the 27 MC circuit is not selected, communication on this circuit will not be heard unless the 27 MC monitor switch (S1908) is on, or the "conn report" or torpedo detection circuit has been actuated. Actuation of the conn report or torpedo detection circuit will energize relay K1931, which connects the 27 MC monitoring isolation resistor R1917 to the monitor volume control and disconnects all other monitoring circuits to prevent interference with the message. Operation of the 31 MC circuit is similar to intercom operation specified in 3.2.12.10.12. The 31 MC switch S1910H grounds the intercom call-in line and operates K1933, applying power to station amplifiers. Audio circuits are connected to the amplifier control in normal manner. Operation of amplifier control talks switch grounds the 31 MC control line permitting communication to the escape trunk loudspeaker in conjunction with the 31 MC control box. When the 31 MC circuit switch is not selected, transmission on this circuit will not be heard unless 31 MC monitor switch S1904 has been operated to connect the audio circuit through the normally closed contact of the selector switch, with the isolation monitoring resistor R1912. Monitor switch S1904 also connects power for the amplifier.

3.2.12.10.11 Periscope - muting. - The selector switch positions just discussed have an additional control circuit which may be used to provide a loudspeaker muting control when adjacent stations selecting the same circuit for communication experience acoustical feedback. These control circuits terminate at terminals 127, 128, 129, and 130 where they are connected by ship's wiring to terminals in the control cabinet, at which point they may be cross connected, as required, to corresponding circuits in the interfering station. Each of these mute control circuits is connected with a normally open contact on the corresponding selector switch. Starting with S1910A, the movable arm is connected through a series of closed contacts on S1910B, S1910C, and S1910H with terminal 188/198 and with a set of contacts on K1925. Terminal 188/198 is externally connected to terminal 42 of the amplifier control which connects to the relay coil of the muting relay K2104. The mutually interfering station has a similar series of connections on its selector switch and to its amplifier control unit. Taking S1910A as an example, the closure of this switch will complete the muting connections from terminal 127 to the relay arm of K1925. K1925 is actuated, as has been noted, whenever talk switch at the periscope station is operated. The control circuit is now completed to terminal 127 and to terminal 128. A set of contacts on muting relay K2104 in each of the affected amplifier control units will be so connected in the loudspeaker circuit, that whenever relay K2104 is energized, LS2101 will be prevented from operating. With a similar connection at the interfering station, operation of K1925 through the other selector switch will energize relay K2104 in the other station. With its loudspeaker muted, acoustical feedback cannot occur. The operation of relay K2104, in the periscope amplifier control box, is not significant since its loudspeaker has already been disconnected from the output circuit either by operation of the press-to-talk switch or by operation of the audio transfer relay. Likewise, when the other station transmits, a control voltage is also connected through terminal 127, through the selector switch contact to terminal 188/198, and to the muting relay in the amplifier control unit. If either station selects a different circuit the interconnection between the two stations is interrupted and the muting relay does not operate.

3.2.12.10.12 Periscope - intercom operation. - Communication on an intercom circuit is accomplished by selecting the desired station. Operation of one of the station selector switches connects the selected

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audio circuit to the selector switch audio buss. The selected station control circuit is interconnected with Conn LO and with relay K1933. Amplifier power relay K1933 is now energized through the control circuit. The corresponding amplifier power relay in the called station is also energized. Amplifier power in both stations has now been applied, and both stations are in a ready condition. Closure of the talk switch and speech into a loudspeaker or microphone at one station will be reproduced at the other station. If the called station has its selector switch in a stand-by position, no other selection is necessary for communication; it is only necessary to operate the talk switch to transfer the station amplifier from a listen to a talk condition. When the intercom communication is completed, the amplifier power at both stations should be turned off by selecting the stand-by position or one of the numbered circuit positions. With relay K1926 in its de-energized position, the Conn intercom audio line is connected to the selector switch audio buss, and to the amplifier control audio circuits. Thus, when the periscope station selects an intercom position for operation, the transmitted and received audio will be present on the audio circuits of both stations. If a third station selects either of these stations, it will be evident that the stations are busy and in communication. This also provides a means for conference operation of several stations. This is accomplished by all stations selecting a common station. Failure to release an intercom selection at the conclusion of communication may result in unwanted conference operation. With the selector switch in the stand-by position, all other selections will be released and power relay K1933 will be de-energized, turning off the amplifier power. Selection of the Conn station from one of the other intercom selectors will energize the station power relay K1933 by way of the Conn intercom call-in control circuit, and power will again be applied to the amplifier to make the station responsive to the incoming communication. To reply, it is only necessary to operate a station talk switch. If one of the numbered circuits has been selected, the station power relay will be energized as before, but the incoming audio signal has now been transferred to the monitor system and will be heard on the station monitor loudspeaker. To reply to this call, it will be necessary either to select the calling station or to select the stand-by position.

3.2.12.10.13 Periscope - radio operation. - The periscope station control box is equipped with facilities for the independent control and monitoring of a radio telephone circuit. Jack J1901 is provided for the connection of a standard Navy handset and is wired to terminals 201/211 through 205/215. A "carrier on" light DS1903 is connected through series dropping resistor R1906 to terminals 206/216 and 207/217. A 6-volt lamp is installed in the receptacle and, with the dropping resistor, the "carrier on" circuit operates on 12 volts. Terminals 208/218 and 209/219 provide for a 600-ohm radio loudspeaker input circuit connected to the loudspeaker volume control R1904. The volume control and an isolation resistor R1925 are connected by way of terminals 85/95 and 86/96 to the audio input terminals of the radio monitor loudspeaker. Amplifier power connection for the monitor loudspeaker is made to terminals 87/97 and 88/98. Terminal 88/98 is the ground connection. Terminal 87/97 is connected to monitor control switch S1905 whose function is to control the power to the radio monitor loudspeaker, thus providing a means for turning it on and off as required.

3.2.12.11 Intercom Set Control C-3056/WIC. - The sonar break-in control box provides the facilities for controlling the communications to and from a maximum of six sonar operators. It is used in conjunction with the evaluator's amplifier control (C-3224/WIC). Two communication circuits are provided: an operator's circuit and the 27MC circuit. In the split mode, the sonar operators may communicate with the evaluator only. In the combination mode the two circuits are interconnected and the operators can communicate on the entire 27 MC circuit.

3.2.12.11.1 Detailed function of sonar break-in control box. - Refer to figure 1, sheet number 5, (Sonar Break-In Box). Each of the operators' control circuits function in the same manner. Operator No. 1 will be used as the example in the following description. Operators No. 2 and 3 are identical to No. 1. Operator No. 4 has provisions for two independent sonar audio inputs, one for each earphone. Operators No. 5 and 6 have no provisions for sonar audio input.

3.2.12.11.2 Operator No. 1 - listen (split mode). - Coupling transformer T3607 is connected to terminals 22 and 23 for external connection to the sonar set audio. One side of the 10-ohm winding of this transformer is connected to terminal 21 for the common earphone connection; the other side is connected to terminal 26 for one earphone connection, and through a normally closed contact of K3603 to terminal 25 for the other earphone connection. When relay K3603 is in its unenergized position, both headset earphones will be connected in parallel across the 10-ohm winding of T3607 and coupled to the sonar set audio signal. The 150-ohm output of line amplifier A3601 is connected to coupling transformer T3601 through relay K3602. One side of the secondary of T3601 is connected to the common earphone connection of T3607, and the other side through a normally open contact on K3603 to terminal 24. Terminal 24 is connected through a volume control in the operator's control box to one of the operator earphones. When relay K3603 is energized, the circuit connection between the 10-ohm secondary of T3607 and one of the earphones is interrupted, and

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instead a connection is made from the 8-ohm winding of T3601 through terminal 24 and the operator's box to this earphone. Thus, the sonar audio is interrupted in this one earphone and the audio output of the line amplifier substituted. Break-in relay K3603 will be energized only when break-in switch S3601 is on. Closing of S3601 completes a control circuit from terminal 65/75 through a normally closed contact on relay K3601 to a normally closed contact on relay K3605. When these relays are de-energized and the operator control circuit is energized by the operation of the evaluator's station on the operator's circuit, relay K3603 will also be energized causing the break-in function to occur. The audio input connection to the break-in box line amplifier is similar to that used in the amplifier control unit except that no volume control is provided. Bridging resistor R3607 and attenuation resistors R3605 and R3606 are connected to the amplifier input. The control of the volume for the operator's earphones is accomplished by the individual volume controls on the operator's control box.

3.2.12.11.3 Operator No. 1 - talk (split mode). - When operator No. 1 operates his talk switch, relays K3604 and K3605 are energized. Relay K3604 connects operator No. 1 microphone circuit to the input of the line amplifier. Relay K3605 energizes the break-in control relay K3603 for sidetone to assist the operator in controlling his volume level. K3605 also energizes K3601 and K3602. These two relays are energized whenever any operator talks out. Relay K3601 connects the microphone preamplifier output to the input circuit of the 1-watt amplifier. Relay K3601 also closes the control circuit to terminal 65/75, and breaks the connection between the operator control circuit and other break-in relays. Relay K3602 connects the 8-ohm output of the line amplifier to the AGC circuit to provide a source of voltage for automatic gain control. Relay K3602 connects the 150-ohm output circuit of the line amplifier to terminal 61/71 for connection to the evaluator's station. Relay K3602 also removes the short circuit on resistor R3609, placing this resistor in series with the common connection to transformers T3601 through T3606. This connection provides some attenuation of the audio signal to the headset so that side-tone will be received at the earphone at a slightly reduced level than the normal communication circuit. When a transmission is made, the operator's control circuit energizes the power relay K3621 in the break-in box and applies +28 volts to line amplifier A3601. The evaluator's amplifier is continuously energized. Thus speech into an operator's microphone will be amplified by line amplifier A3601 and transmitted to the evaluator's station. Other operators, however, will not receive the communication even though they may be connected for break-in with respect to communication from the supervisor.

3.2.12.11.4 Operator communication (combined mode). - If it is desired that an operator(s) communicate on the 27 MC circuit mode, switch S3610 is operated to the COMB position. In this position, the spare relay (K2104) in the evaluator's amplifier control is energized. This relay interconnects the operator's audio and control circuits with the system 27 MC audio and control circuits. Indicator lamp, DS3602, will be let to indicate the selection of the combined mode. Communication by the operators will be as above, except that operation will energize the entire 27 MC circuit.

3.2.12.11.5 Operator talk (Conn Report). - If an operator desires to make a report to Conn, the CONN REPT switch is operated. It is not necessary to operate the foot switch or hand-held talk switch on this operator. One section of the CONN REPT switch completes a circuit between ground or LO the operator's talk control circuit. The other section of the CONN REPT switch closes through the connection between the ground or LO and Conn Report control circuit, which is externally connected to terminal 68/78 in the sonar break-in control box. The operator Conn Report circuit in turn energizes the system conn report circuit and the circuit combining relay K2104 in the evaluator's amplifier control unit (see 3.2.12.11.4). In this manner the operator will be temporarily connected to the 27 MC circuit during the Conn Report. The Conn Report circuit energizes a special relay in the periscope/bridge control box so that the operator's report will be transmitted to the Conning position.

3.2.12.11.6 Operator talk (torpedo detection). - Operation of the TORPEDO DET switch also completes the circuit between ground and the talk control, and in addition, closes a control circuit which is externally connected through junction point 10/20 in the operator break-in box to the control cabinet. At the central control cabinet, an audio circuit to the Forward and After Signal Ejector stations is energized so that the loadspeakers at these stations will receive any announcements made on this circuit. In addition, the operation of the Torpedo Detection circuit energizes the Conn Report circuit (so that the Conning position receives a Torpedo Detection report) and the circuit combining relay (see 3.2.12.11.4).

3.2.12.11.7 Evaluator Communication. - In the split mode the evaluator may select either circuit for communication (see 3.2.12.4). The unselected circuit is automatically monitored to insure receipt of incoming messages. In the combined mode the position of the selector switch is inmaterial; the evaluafor talks on the combined circuit.

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3.2.12.11.8 Evaluator (Conn Report and Torpedo Detection). - The Conn Report and Torpedo Detection switches on the break-in box, for the evaluator operate as described above except that the evaluator's amplifier control talk circuit is energized.

3.2.12.11.9 Sonar monitor. - The evaluator sonar audio monitoring facility is provided by an additional winding on transformers T3607 through T3611. These windings are connected to different positions of the sonar monitor switch, for example on T3607 through S3607A-3 and 1 and K3621-3 and 1, and isolation resistor R3604 to monitor volume control R3601, and thence through isolation resistor R3603 to terminal 64/74. At terminal 64/74, an external connection is made to the monitor output terminal of the evaluator's amplifier control box."

3.2.12.11.10 The sonar operator interconnecting boxes, J-1070/WIC and J-1071/WIC. - (Refer to figure 1 sheet 5.) These boxes serve as a junction and control point between the operator's headset and microphone and the break-in control box. Two similar types of boxes are to be used; the first type is to be provided with torpedo detection reporting circuit and the Conn Report circuit. The second type of box will not include the torpedo detection reporting circuit. Connected to the terminal board inside the case of the operator's interconnecting box is a length of flexible cable (CX-4692/WIC) which is terminated in a special connector to accommodate the split circuit headset (H-179/U) provided for the operator's use. In the overall schematic diagram, a foot switch is shown connected to J3703, but this may instead be a hand-held button-operated switch. The operation of components of the sonar operator's interconnecting boxes is covered in the description of the break-in box.

3.2.12.12 Control Cabinet CY-2671/WIC. - This is a control cabinet which is to provide for the central interconnecting point for all the communication facilities except those associated with the 21 MC intercom circuit.

3.2.12.12.1 Detailed function of circuit cutout and relay panel. - Refer to figure 1, sheet no. 1 of the overall schematic diagram. The uppermost panel contains the disconnect switches for the intercommunication stations, the general announcing loudspeaker cutout switches, the emergency reports cutout switches, the system priority relays, the emergency reports system line amplifier, the radio monitoring system isolation amplifier, and power circuit fuses. Each fuseholder has an integral incandescent bulb wired across the fuse to provide a blown fuse indication. Rotary wafer switches S101 through S116 provide a means for disconnecting leads to a station area except the ground lead. Switches S112, S113, and S115 are spare switches and are not interconnected or wired to terminal boards. Toggle switches S117 and S136 provide for the disconnection of 20 emergency reports circuits. Toggle switches S137 through S166 provide for the disconnection of 30 general announcing reproducer groups. Group 20 is connected to provide for a 1 MC/4 MC loudspeaker. This loudspeaker, located in the maneuvering room, supplies 4 MC transmission and assures receipt of 1 MC transmission in the maneuvering room. The last five groups, G26 to G30 inclusive are connected to provide a 1 MC/2MC loudspeaker circuit. The loudspeakers in the engine room and auxiliary machinery area connected to these switches and normally operate on the 2 MC audio circuit, but are transferred to the 1 MC audio circuit for transmission of general announcements or alarm signals whenever the maneuvering room station transmits on 2 MC. Located on the relay panel are the 150-ohm load resistors R108 to R115 for each of the audio distribution circuits. Also included on the relay panel are the isolation resistors and volume controls for the radio monitor isolation amplifier and for the emergency reports audio amplifier. The 1 MC control and priority circuits function as follows: The Conn station (periscope or bridge) has the highest priority on 1 MC voice. This is subject to the higher priority of an alarm signal or emergency report. The low side of control circuits for the periscope/bridge station are connected through terminal 21 to the disconnect switch S101C. S101C is connected through relay K102 to terminal 360, Conn LO, which is grounded in the power distribution panel by K1205. If an alarm signal causes relay K1205 to operate or an emergency reports operation causes relay K102 to operate, this control circuit to the conning station will interrupt operation at the conning station. When the bridge/periscope station operates on 1 MC, the control circuit is closed through to terminal 13 and then through disconnect switch S101B to relay coil K107. The Conn LO circuit at relay K102 is also connected through the normally closed contact of relay K107 and disconnect switch S102A to terminal 55 which is the diving officer station LO for control circuits. The operation of the 1 MC circuit at the conning station will energize relay K107 which interrupts the diving officer LO, thus preventing the operation of that station. The operation of relay K107 by the conning station energizes a 1 MC control circuit, energizing relay K108 and, if the maneuvering station is not talking on 2 MC, K111. Relay K108 removes the ground from the 2 MC LO circuit to prevent the operation of 2 MC stations during 1 MC operation. Relay K108 energizes the 1 MC master circuit connected to terminal 349 and also energizes relay K106. Relay K111 transfers the 1 MC/2 MC group

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of loudspeakers from their normal connection on the 2 MC circuit onto the 1 MC audio circuit. Relay K106 removes the ground connection from the 7 MC LO circuit and prevents the operation of talk back stations on this circuit during 1 MC operation. Relay K106 also grounds the 7 MC control circuit connected to terminal 351 which has no function in 1 MC operation, other than to illuminate the station busy lights indicating that the system is in operation. The 1 MC master circuit at terminal 349 and the 7 MC control circuit at terminal 351, both of which have been energized by the 1 MC operation, are connected by control wiring to the power distribution panel where they energize the group 28 volt power circuits for powering the loudspeaker and station amplifiers required to receive the 1 MC signal or announcements. When either the conning station or the ballast control panel talk on the 1 MC circuit, the audio output is connected to the control cabinet on terminal 1. After passing through the station disconnect switches, the 1 MC audio circuit is connected by K102 with the 1 MC audio circuit for other stations and distribution to the general announcing loudspeaker groups. If a transmission is made by the maneuvering station on the 2 MC circuit, a control circuit is closed through to relay K109. Relay K109 interrupts 2 MC LO and prevents operation of K111. Since K111 cannot operate, the 1 MC/2 MC group of loudspeakers will remain connected to the 2 MC audio circuit even though the 1 MC circuit is in use as specified above. The external contactors or switches for energizing the alarm signals are connected to terminals 339 to 343 inclusive. Terminals 344, 355 and 356 supply the ground or common side of the circuit. The alarm signal control terminals are interconnected by control rack wiring to the signal generator assemblies. When any alarm signal circuit is energized, the action of the control relays in the alarm signal generator energizes the alarm master circuit connected to terminal 348. Terminal 348 is interconnected by rack wiring to the distribution panel where it energizes the master control relay supplying the amplifier power to the system. Terminal 348 is also connected to relay coil K110 on the relay panel. Relay K110, when energized by the alarm master control circuit, energizes relays K108 and K111 by interconnecting the 1 MC control circuit with the 1 MC master circuit on terminal 349, which is energized by the action of the alarm master relays in the distribution panel. The control panel relays now function for circuit control and priority as described for 1 MC voice and the alarm signal generator audio output appearing on terminals 337 and 338 is connected to the 1 MC audio line and the audio common line respectively, and thus distributed throughout the system in the same manner as the 1 MC voice signals. Relays K101 and K102 are the 4 MC emergency reports input and circuit control relays. One of the functions of K102, that of controlling the priority of emergency reports over the Conning station operation, has already been described. The other function of relay K102 is to disconnect the Conning, Diving Officer stations and manuvering 1 MC audio circuit from the normal 1 MC audio distribution circuit, and instead to connect these two stations to the output of the emergency reports amplifier A101. Thus, when relay K102 is energized by an emergency report operation, the loudspeakers at the Conning and Diving Officer stations are disconnected from the normal 1 MC circuit and will receive instead the output of an emergency reports amplifier. The balance of the 1 MC audio distribution circuit is left undisturbed so that simultaneous transmission of alarm signals may be made to other stations in the vessel. Relay K101 closes through the audio circuit from the emergency reports disconnect switches to amplifier A101 through bridging resistor R105 and volume control R101. Relays K104 and K105 are associated with the torpedo detection circuit. These two relays are connected in parallel and through disconnect switch S104 to terminal 79. Terminal 79 is connected through the sonar break-in box to the sonar operator control box having the TORPEDO DET switch. When this circuit is energized by an operator, relays K104 and K105 will be energized. At relay K104, the Conn report control circuit to the periscope and bridge station is disconnected from the normal Conn report control line and connected directly to ground in order to energize the Conn reporting relay in the periscope station. The other set of contacts on K104 connects the signal ejector audio circuit to the 27 MC audio circuit. The signal ejector audio circuit is connected through disconnect circuits S106 and S108 to terminals 207 and 136 respectively. These terminals are connected to loudspeakers located at the forward and after signal ejector stations. Relay K105, when energized, interrupts the 27 MC LO circuit so as to prevent the operation of other stations on the circuit during the time of the torpedo detection report. The operation of the TORPEDO DET switch by the operator will have energized his communication circuit, and speech into his microphone will be amplified and distributed over the 27 MC audio circuit. All stations normally connected to this circuit will hear the transmission, and in addition, the Conning station will hear the transmission as well as personnel at the signal ejector stations. Relay K106 provides for priority of the conning station over operation of other 7 MC stations. In the de-energized position, the ground circuit is connected to the 7 MC LO circuit. In the energized position, the 7 MC LO circuit is interrupted and the ground is transferred to the 7 MC control circuit, which in turn energizes the power relays as previously specified.

3.2.12.12.2 Switch box emergency reports SA-718/WIC. - This is a spring return device used to transfer the sound powered telephone handset from its associated telephone circuit to the emergency reports circuit 4 MC. A typical Emergency Reports switch station is shown connected to terminals 317 through 320. When the emergency reports switch is operated, the sound powered telephone is connected to the audio circuits 317 and 318. At the same time, closure of the switch places a connection between the ground circuit terminal 320 and the control circuit on terminal 319. Audio circuits 317 and 318 are connected through emergency reports group 20-disconnect switch S136 to the emergency reports audio buss, which is controlled

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by relay K101, as previously specified. The control circuit, terminal 319, is connected through group disconnect switch S136 to the control buss, and then to relay coils K101 and K102 to operate the Emergency Reports system. If the group has been disconnected at the control panel, the audio lines to the Emergency Reports station will be short circuited by the disconnect switch in the off position. Thus, when an attempt is made to use the Emergency Reports station, the short circuit will cause the line to appear dead as an indication that a transmission cannot be made. Likewise, if relay K101 should fail to operate due to lack of control voltage, the audio lines will be short circuited, and the system will appear dead when an effort is made to transmit.

3.2.12.12.3 Power distribution panel SB-1056/WIC. - For detailed function of power distribution panel, power supplies, and battery, refer to figure 1, sheet no. 2 of the overall schematic diagram. The power distribution panel provides all the power switching and distribution facilities for the entire AN/WIC system. Located in the electrical equipment cabinet, it provides for (a) power supply and battery selection (b) power supply and battery test (c) emergency battery disconnect (d) failure indication (e) power distribution on selected circuits and (f) miscellaneous alarm indications. These functions are specified in detail as follows:

3.2.12.12.3.1 Alternating current power (115 volts) is obtained from the I.C. Switchboard and connected to terminals No. 514 and No. 518 of the electrical equipment cabinet. Alternating current power ground is connected to terminal No. 518. Terminal No. 515 is connected to the ship's hull. Terminals No. 513, 515, and 518 are connected together and to terminal No. 479 and other ground terminals. Connections from this terminal board to the power distribution panel are by means of a cable and plug P103.

3.2.12.12.3.2 Alternating current power enters the panel through J1205 and is connected to the power supply selector switch S1201. Each wafer of this switch is a simple SPST switch; in position No. 1, sections A, C, E, and G are closed; in position No. 2 sections B, D, F, and H are closed. Therefore, in position No. 1, the a.c. power is connected to J1201 through fuses F1201A and B. In position No. 2, the a.c. is connected to J1202 through fuses F1202A and B. Cables W101 and W102 are connected to the two power supplies (numbered 1 and 2) and supply the a.c. power. It will be noted that a.c. is connected to only one power supply, hereafter designated "active". The other power supply is designated as the "standby" supply.

3.2.12.12.3.3 The d.c. output of power supply No. 1 is also connected to J1201. Power supply No. 2 is connected to J1202. The negative leads of the power supplies and battery are connected together and connected through J1204 to cabinet terminal No. 479, which is the system ground. All switching is done to the positive connections. The d.c. from the active supply is connected to the anode of a blocking diode CR1201. This diode prevents battery discharge through a defective power supply. The cathode of CR1201 is connected through the NC contacts of S1203 and S1206F (normally closed) to the storage battery connected to J1203.

3.2.12.12.3.4 The connection between S1203-4 and S1206-F is the junction point of the active power supply and the battery. The following circuits are connected to this point: (1) the low current (20A) circuit through F1203 to the distribution relays; (2) the high current (30A) circuit through S1206H (normally closed) and F1204 to the 1 MC power relay K1218; (3) the power supply failure indcators through the supervisory relays K1201, K1202 and K1203.

3.2.12.12.3.5 The storage battery is used in the system for two reasons: (1) to supply additional current to the system during infrequent peak-demand periods and (2) supply all system power in the event of rectifier or a.c. power failure. To accomplish this, the battery is floated across the system d.c. supply. The power supply voltage is approximately 28 volts which keeps the 24 volt battery fully charged. The regulation of the power supply causes its voltage to drop as more current is required. As the power supply voltage drops, the battery supplies additional current. In the event of power supply failure, the battery is already across the line and supplies all the current. Discharge through the defective power supply is prevented by blocking diode CR1201.

3.2.12.12.3.6 In order to properly set up the system voltage, the active power supply shall be adjusted under open-circuit conditions to eliminate the battery influence and the normal system current drain. This is accomplished by operating toggle switch S1203 momentarily. The voltmeter (M1202) is normally connected between ground and system voltage by means of S1203-3 and S1204-3 and 1. Operation of S1203 connects the voltmeter to the power supply by means of S1203-3 and 5. During this test, the system will continue to function since the battery is still on the line. There will be no failure indication since the power supply continues to energize K1201. While S1203 is operated, an additional load R1202 and R1203 is placed on the system so that the battery will be partially discharged. Battery discharge may be observed on the ammeter

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(M1201) with its shunt R1202. Upon release of S1203, the system returns to normal and the battery should quickly regain its full charge as shown by the ammeter.

3.2.12.12.3.7 The stand-by power supply may be tested in the following manner: Alternating current power input is obtained by interconnecting the line side of fuses F1201A and B and F1202A and B by operation of switch S1205. The d.c. output is connected through S1201-G or H to S1204-6 or 5. Operation of S1204 connects the standby power supply to the voltmeter by means of S1203-1 and 3.

3.2.12.12.3.8 S1206 provides a means for disconnection of the battery circuits in the event of battery failure. Operation of this switch provides the following operations: sections A and C interconnect the line side of fuses F1201A and B and F1202A and B; section F disconnects the battery; section G connects the "stand-by" power supply to the high current (30A) line which has been disconnected from the "active" power supply by section H. The voltmeter (M1202) will now indicate the "stand-by" power supply voltage unless S1203 is operated.

3.2.12.12.3.9 The energized coil of relay K1201 is connected between ground and the "active" power supply by means of its own NO contacts 3 and 5. In the event of power supply failure, voltage will be removed from the coil of K1201 causing it to release and connect power signal line to the system voltage through K1203-4 and 6, K1202-4 and 6, and K1201-6 and 8. The power supply signal line is connected to terminal No. 478 in the electrical equipment cabinet, which is connected via terminal No. 61 to the power supply failure alarm located in the control room. The diode (CR1201) also prevents the battery from energizing relay K1201. The lamp (DS1203) connected between K1201-6 and ground is also energized. When the source of the power supply failure has been corrected, operation of the reset button (S1202) will energize the coil of relay K1201 temporarily so that contacts 3 and 5 may close and hold the relay on.

3.2.12.12.3.10 The coils of the other two relays (K1212-2 and K1213-2) are connected from the left side of fuses F1203 and F1204 to ground. Should either fuse fail, then voltage would be removed from the coil of the respective relay causing it to release and connect the power supply signal line to the system voltage. Meanwhile the respective lamp (DS1204 and DS1205) will be lit indicating the blown fuse. Replacement of the fuse will automatically reset the alarm system.

3.2.12.12.3.11 The 28 volt power is distributed on nine lines which connect through J1204 to terminals on the electrical equipment cabinet: one line (No. 480) is always connected to F1203. The remaining eight lines (Nos. 469 through 477) are "Group + " lines. One side on the coil of each relay K1204 through K1216 are relays K1219, K1220, and K1221, are connected to F1203 via TB1201. The "Group+ " lines are controlled by seven control lines connected to terminals of the electrical equipment cabinet. Grounding any of the control lines will cause +28V system power to be connected to certain group + lines as indicated in table II. The relay operation is as follows:

Table II - Control of group power circuits.

	Alarm Master 348	1 MC Cont. 349	2 MC Cont. 350	7 MC Cont. 351	27 MC Cont. 352	32 MC Cont. 353
1 MC + 469	x	x				
1 MC + 470	x	x				
2 MC + 471	x	x	x			
7 MC + 472	x	x		x		
27 MC + 473					x	
Eng + 475	x	x	x	x		
Miss. + 476	x	x		x		x
Conn + 477	x	x	x	x	x	x

3.2.12.12.3.12 The 1 MC master control line (349) is grounded by external circuits during 1 MC voice operation and by K1206 during alarm operation. It will cause K1204, K1207, and K1219 to operate. Contacts 3 and 5 of K1204 ground the 7 MC control line (351). Contacts 6 and 4 of K1204 ground the 2 MC control line (350). Operation of these control lines is detailed in following paragraphs. Contacts 3 and 5 of K1207 close the S.E. Mute (1 MC) control circuit connected to terminals 502 and 504. K1219 connects 1 MC + A (469) and 1 MC + B (470) to system power through F1204.

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3.2.12.12.3.13 The 2 MC control line (350) is grounded by operation of the 2 MC announcing circuit or by operation of K1204 (1 MC master). It operates K1212 and K1213. Relay K1212-3 connects the 2 MC - line (471) to system power (pin 5) from TB1201. Relay K1213-6 connects the ENGR + line (475) to system power (pin 4) from TB1201. Contacts 3 and 5 of K1213 ground K1208-7 and cause it to operate. Relay K1208-3 connects the CONN + line (via K1209-6 and 8) to system power (pin 5) from TB1201.

3.2.12.12.3.14 The 7 MC control line (351) is grounded by operation of the 7 MC announcing circuit or by operation of K1204 (1 MC Master). It operates K1209, K1210, and K1211. Relay K1209-6 connects the CONN - line (477) to system power (pin 4) TB 1201. Relay K1210-6 connects the MISS + line (via K1215-6 and 8) to system power (pin 4) TB 1201. Relay K1211-3 connects the 7 MC - line (472) to system power (pin 5) TB1201.

3.2.12.12.3.15 The 27 MC control line (352) is grounded by operation of the 27 MC announcing circuit and operates K1214. Relay K1214-6 connects the 27 MC - line (473) to system power (pin 4) from TB-1202. Contacts 3 and 5 of K1214 ground K1208-7 causing it to operate and connect system power to CONN + (477).

3.2.12.12.3.16 The 32 MC control line (353) is grounded by operation of the 32 MC announcing circuit and operates K1215. Relay K1215-6 connects the MISS - line (476) to system power (pin 4) TB1201. Contacts 3 and 5 of K1215 ground K1208-7 causing it to operate and connect system power to CONN + (477).

3.2.12.12.3.17 The alarm master control line (348) is grounded during the operation of any of the alarm signals generated by the alarm signal generator. This line is connected to K1205-7 and K1206-7, and will cause these two relays to operate. Conn LO (360) is connected to ground through K1205-8 and 6 and K1205-1 and 3 in parallel. Operation of K1205 opens this circuit and prevents CONN operation. Contacts 3 and 5 of K1206 close the S.E. Mute (alarm) control circuit connected to terminals 501 and 503 respectively. This will cause the S.E. system to become inoperative during alarm signals. Contacts 6 and 4 of K1206 connect the 1 MC Master control line (349) to ground causing all circuits to become activated except 27 MC + (473).

3.2.12.12.3.18 A contactor is provided for a.c. switching during alarm operation. Designated K1221, it is energized by visual alarm control (346) from the alarm signal generator. When energized, alternating current power input to terminals 506 and 508 will be connected to terminals 510 and 512, respectively, for visual alarm indication.

3.2.12.12.4 Power Supply Signal BZ-55/WIC. - Refer to figure 1, sheet No. 4. The power supply failure alarm box has both a buzzer (DS3101) and an indicating lamp (DS3102). The lamp and buzzer will indicate if terminal No. 1 is connected to positive power. The buzzer may be switched off by operating S3101, which connects the coil of K3101, to the positive signal line. Contacts 6 and 4 and 3 and 5 of K3101 close and keep the relay energized. The buzzer (DS3101) is turned off, since relay contacts 8 and 6 and 1 and 3 are open. When the positive voltage signal is removed from terminal No. 1, then the relay (K2101) and lamp (DS3102) will become unenergized. Therefore the buzzer (DS3101) will sound the next time failure occurs.

3.2.12.12.5 Electrical Test Panel SB-991/WIC. - For detailed function of test panel (Refer to figure 1, sheet No. 2 of the overall schematic diagram). The electrical test panel contains the alarm signal generators and facilities for testing amplifiers and relays of the AN/WIC system. The panel is located in the electrical equipment cabinet.

3.2.12.12.5.1 A cable and plug (P101) attached to the cabinet terminal boards makes all connections to J1308 of the test panel. Plug P101 is arranged so that it may be plugged directly into a signal generator assembly (OA-2176(v)/WIC-1) in event of failure of the electrical test panel.

3.2.12.12.5.2 The alarm signals are generated by one of the two operating-assemblies (OA-2176(v)/WIC-1) located in the panel. Selection of the generator assembly to be placed "ON LINE" is made by S1301. The other signal generator is automatically placed in test position. Section A of S1301 illustrates the transfer operation. When S1301 selects No. 1 ON LINE, audio out of a signal generator No. 1 (A1301) is connected through J1301-7 to J1308-7 via S1301A-3 and 4. Audio out of signal generator No. 2 (A1302) is connected from J1302-7 to the test circuits by means of S1301A and -1. When S1301 is reversed, audio from signal generator No. 2 (J1302-7) is connected to J1308-7 via S1301A-2 and 3. Signal generator No. 1 audio is connected to test via S1301A-4 and 5. Visual alarm control, switch No. 2, switch No. 1 horn control, switch

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No. 3 switch No. 5, switch No. 4, alarm master switch No. 7 and switch No. 5 are transferred in a similar manner by S1301, sections A through F.

3.2.12.12.5.3 The control switch leads from the alarm contactors are connected to cabinet terminals Nos. 339 through 343, 358 and 359. Connections are then made through P101 and J1308 pins 1, 2, 8, 9, 4, 15, and 16 and S1301 to the active signal generator assembly. Audio output of the active signal generator is connected to the 1 MC audio line in the cabinet by means of S1301A through pin 7 of J1308 and P102 and terminal No. 337. Audio common of both signal generator assemblies are connected to E1302 which is connected through J1308 and P102 pin 14 to terminal No. 338. The three control circuits of the active signal generator are connected by S1301 to power distribution panel (SB-1056/WIC) via J1308, P101 and cabinet terminals 346, 347, and 348. Power (+28 V) is connected from terminal No. 345 to the active signal generator via P101 and J1308, pins 5 and 12, and S1301E-9 to the active signal generator assembly pin 12. Operation of any external control switch lead will also cause K1302 to operate. Operation of K1302 applies power to the control circuits of the active signal generator assembly pin 5, and connects the alarm master control from the active signal generator to J1308-11. S1301E also connects power to the control circuits of the generator assembly in test.

3.2.12.12.5.4 Power (+28 V) for the stand-by signal generator and other test circuits is controlled S1308. Fuse F1301 protects the system supply from a possible fault in a unit being tested. An indicator lamp DS1301 indicates when test power is ON.

3.2.12.12.5.5 The seven switch lines from the stand-by signal generator are connected by S1301 to test switches S1302 through S1306, S1309, and S1310. These switches control the stand-by generator in the same manner as the ship's alarm contactors control the active signal generator. Operation of the three control lines by the stand-by signal generator is indicated by the three panel lamps DS1302, DS1303 and DS1304. Audio output is developed across load resistor R1301.

3.2.12.12.6 Audio test facilities. - Facilities are provided for conducting an audio gain test of the signal generator assembly (OA-2176(v)/WIC, the high power amplifier (AM-2118A/WIC) and the amplifier assembly (AM-2158A/WIC) in both talk and monitor conditions. Selection of the unit to be tested is made by S1307. Detailed operation of each position is as follows:

3.2.12.12.6.1 The panel meter (M1301) will indicate "O DB" when placed across a 600 ohm, one milliwatt circuit (i.e. 0.77V). Each of the test circuits should develop one watt across 150 ohms or 12.2 volts. A voltage divider (R1303 and R1305) is provided to reduce the 12.2 volts signal to 0.77 volt required for the meter.

3.2.12.12.6.2 The primary of T1301 is placed across the meter circuit so that the signal may be monitored. The secondary is connected to headset plug (J1303D and E). The impedance ratio of T1301 is 1500:8 which will develop 100 milliwatts in the headset with little system loading.

3.2.12.12.6.3 In the ALARM GEN position, the audio output of the stand-by signal generator is connected directly to the meter circuit through S1307B-2 and 1. The signal generator assembly output level is set to "O DB" for the power plant failure alarm (No. 4). Adjustment is made by R902 on the signal generator assembly.

3.2.12.12.6.4 In the AMPL HI-PWR position and with the test signal adjusted, to "O DB", 12.2V will appear across R1301 when No. 4 alarm is operated. The isolation step-down transformer (T1302) will then produce 0.75 volts across R1302. The low side of all audio test circuits connect to a special audio common independent of system audio common. Meter low is connected to this common in amplifier test positions by S1307B-7 and 9, 10, 11, and 12. The audio signal from R1302 is connected through S1307A-7 and 9 to series isolation resistor R1306. The high power amplifier (AM-2118A/WIC) is contained in a special test fixture which plugs into the test panel. An integral cable makes the necessary connections between J1311 (for the amplifier plug P401) and P1301 which plugs into J1304 on the test panel. The same pin number is used on each connector for a given circuit. Input from R1306 is made through pin 2 to the arm of the volume control mounted on the amplifier. Volume control high (pin 8) is connected to input (pin 1) at P1301. Power is supplied to pins 5 and 12 from F1301 via S1307C-1 and 3. Output is taken from the 250-ohm winding (pin 14) and common out (pin 6) which is connected to test audio common. Output load is provided by R1307 and R1308-R1309 in parallel. Voltage divider R1308 and R1309 is set to develop 12.2 V across R1309 for an amplifier with nominal gain. The amplifier is operated at 7.5 watts output (43 V) to reduce interference between the class B current drawn by the amplifier and the signal generator on the same power line. The audio output across R1309 is connected through S1307B-3 and 1 to the meter circuit.

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3.2.12.12.6.5 In the DISTR AMP-MON position, the audio signal from R1302 is connected through S1307A-7 and 10, and a series isolation resistor R1310, to the transformer input of the AM-2158A/WIC through pin 10 of J1305 and P1001. R1311 stimulates the volume control loading in the monitor condition. The transformer output (pin 5) is connected to the one watt amplifier input (pin 4) through S1307A-4 and 1. Power is supplied for all distribution amplifier tests to pin 12 through S1307C-1 and 4, 5, and 6. Amplifier output for all tests is taken from the 150 ohm winding (pin 7) through S1307B-1 and 4, 5, and 6 to the meter circuit. Loading is provided by R1315. The eight ohm output (pin 6) is always connected to the AGC input (pin 11). The input is set to give one watt output and a "O DB" reading.

3.2.12.12.6.6 In the DISTR AMPL-TALK-HI position, the audio signal from R1302 is connected through S1307A-7 and 11, and a series resistor R1312, to the five ohm input of the AM-2158A/WIC through K1201-8 and 6 and J1305-P1001-1. R1314 provides the proper source impedance. The preamplifier output (pin 3) is connected to the 1-watt amplifier input (pin 4) through S1307A-5 and 1. Power and output switching are as specified in 3.2.12.12.6.5. This input is set approximately 10 db above breakpoint (+1 db on meter) and the indicated output should be less than +4 db.

3.2.12.12.6.7 In the DISTR AMPL-TALK-LO position, the audio signal from R1302 is connected through S1307A-7 and 12, and a series resistor R1313, to the 150 ohm input of the AM-2158A/WIC through K1201-1 and 3 and J1305-P1001-2. Other connections are the same as in the paragraph above. This will give the input (below AGC) necessary to give a nominal "O DB" reading.

3.2.12.12.7 Headset test. - Operation from a headset microphone plugged into J1303 is possible in either HI or LO TALK position. It is necessary to energize K1201 by grounding J1303C to J1303E. This relay opens the signal source from R1314 and closes the five ohm microphone circuit from J1303J to the amplifier (pin 1). The high side of a 150 ohm microphone circuit connected to J1303A is also closed to the amplifier (pin 2) by the K1301-3 and 5. Speech into the amplifier will be indicated on the meter and in the headset.

3.2.12.12.8 Relay test. - A relay may be plugged into J1306 for an operational test. If the NC relay contacts are made, panel lamp DS1305 will light since the circuit is made from P1301 through S1309-3 and 1, J1306-8 and 6, J1306-3 and 1, S1309-2 and 4 and DS1305 to ground. When S1309 is operated, the relay coil (K1306-2) is energized. If the NO relay contacts are made, the lamp will continue to light since the circuit is now made through S1309-3 and 5. J1306-6-4 and 6, J1306-6-3 and 5, S1309-6 and 4 and DS1305 to ground.

3.2.12.13 AM-2319/WIC. - This is a low power loudspeaker comprised of a 1-watt amplifier integrally mounted in a housing in which is mounted a loudspeaker unit.

3.2.12.13.1 Detailed function of the low-power loudspeaker. - (Refer to figure 1, Sheet No. 1.) Audio input at line level (12.2V) is connected to input terminal 3/13 and to audio common, terminal 4/14. The signal passes through bridging resistor R1501 to the arm of potentiometer R1502 which is used to set the operating level. The output of R1502 is connected to terminal 11 which is linked to terminal 1, the input to the amplifier. The loudspeaker is connected between the eight-ohm output of the amplifier and ground. Power is applied to the amplifier by connecting terminal 6/16 to a 28V line. When the unit is used as for monitoring the input is connected to terminal 2/12 since the bridging resistors are located in the station control box.

3.2.12.14 AM-2320/WIC. - This is a high power loudspeaker comprised of a 20-watt amplifier integrally mounted in a housing, in which is mounted a loudspeaker unit.

3.2.12.14.1 Detailed function of the high power loudspeaker. - (Refer to figure 1, Sheet No. 1.) Audio input at line level (12.2V) is connected to input terminal 3/13 and to audio common, terminal 4/14. The signal passes through the bridging resistor R1401 to volume control input pin J1401-P401-2. The level setting potentiometer is part of the 20-watt amplifier. The output of the potentiometer pin P401-J1401-8 is connected to terminal 11 which is linked to terminal 1 which is connected to the amplifier output J1401-P401-1. The outputs of the amplifier (8 ohm, 16 ohm, 250 ohm, and COM) are connected to terminals 8, 9, 10 and 7 respectively. The 16 ohm loudspeaker is also connected to terminals 7 and 9. Power is applied to the amplifier by connecting terminal 6/16 to a 28V line.

3.2.12.15 Modification Kit MK-518/WIC. - This unit is to be used in conjunction with either AM-2319/WIC or AM-2320/WIC loudspeakers to provide external control of the volume between two present levels.

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3.2.12.15.1 Detailed function of the loudspeaker modification kit. - Refer to figure 1, Sheet No. 1, (typical loudspeaker with volume control). The kit may be used for several diverse applications.

3.2.12.15.2 Volume control action. - For this operation the audio line is connected to the volume control input, terminal 18, and to audio common, terminal 4/14. Power (+28V) and ground are connected to terminals 6/16 and 5/15, respectively. The audio output of the volume control is connected to the line input terminal, 3/13, of the loudspeaker. With the relay in the de-energized condition, an audio signal at the volume control is connected through relay contacts K3401-3 and K3401-1 to the loudspeaker amplifier. The level is then set by the loudspeaker potentiometer. When the relay is operated by grounding terminal 17, the audio line is witched through relay contacts K3401-3 and K3401-5, R3401, and R3402 to amplifier input terminal 1/11. R3401 and R3402 now serve as the bridging resistor. R3401 determines the amount of bridging and therefore the output level of the loudspeaker. R3402 limits the maximum input to the loudspeaker amplifier.

3.2.12.15.3 Muting by external control. - Operation in this mode is similar to that described for volume control action (see 3.2.12.15.2), except that the auxiliary output of the modification kit is not connected to terminal 1/11. Therefore when the relay is operated, by grounding terminal 17, audio is removed from the amplifier input.

3.2.12.15.4 Muting by amplifier control. - Muting of the audio circuit is accomplished in the manner specified in 3.2.12.15.3 above. The control circuits, however, are changed as follows: The link between terminals 6 and 16 is removed. Power (+28V) for the loudspeaker amplifier is connected to terminal 6. Both connections of the relay coil (terminals 16 and 17) are then extended to the amplifier control, and connected in parallel with the talk relays in that unit (terminals 70/80 and 28, respectively).

3.2.12.16 *Overall system interconnect schematic drawing.* - The contractor shall be responsible for maintaining accurate overall system drawing and for modification of this drawing in the event that changes to the drawing are authorized by the bureau or agency concerned. When changes are required and authorized, the contractor shall modify and submit the drawing for approval. The drawing shall clearly delineate the changes made and the purpose of the change.

3.3 Figure 1, Sheets 1 through 10, issued under Amendment -1 to Specification MIL-S-23012 dated 24 October 1961, forms a part of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections *set forth in the specification where such inspections are deemed necessary* to assure supplies and services conform to prescribed requirements.

4.2 General requirements. - The sampling procedures for the various tests to be performed on the equipment shall be as specified in 4.4. (The attention of the manufacturer is invited to the applicable paragraphs of Specification MIL-I-983 for information concerning these tests such as where conducted, failures, and test conditions).

4.3 Qualification tests [1/]. - Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Equipment components in the quanitities indicated below shall be assembled as a representative system and subjected to the qualification tests specified herein.

Quantity	Equipment components
2	Intercom Set Control C-3047/WIC
1	Intercom Set Control C-3054/WIC (15 Station Control Box)
1	Electrical Equipment Cabinet CY-2671/WIC-2
1	Amplifier Loudspeaker AM-2319/WIC
1	Amplifier Loudspeaker AM-2320/WIC
1	Periscope Station Type C-3049/WIC
2	Headset Microphone H-178/U
1	Microphone M-106/U

[1/] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

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Test	Requirement paragraph	Test paragraph
General examination	3.2	MIL-I-983
Cross talk	3.2.6.15	4.5.1
Switching transient noise	3.2.6.16	4.5.2
No load output voltage	3.2.7.1	4.5.3
Output voltage regulation	3.2.7.2	4.5.4
Ripple voltage	3.2.7.3	4.5.5
Input power consumption	3.2.7.4	4.5.6
Amplifier test (conditions of)	3.2.8	4.5.7
Frequency response	3.2.8.4.1	4.5.7.3
Distortion	3.2.8.4.1	4.5.7.4
Power output	3.2.8.4.1	4.5.7.5
Automatic gain control	3.2.8.4.1.3.5	4.5.7.6
Rated load current	3.2.8.4.1	4.5.7.7
Cross talk (power line modulation)	3.2.8.4.1.4	4.5.7.8
Alarm signals	3.2.9.1	4.5.8
Mircophone impedance	3.2.11.1.1	4.5.9.1
Microphone frequency response	3.2.11.1.2	4.5.9.2
Microphone sensitivity	3.2.11.1.3	4.5.9.3
Microphone endurance	3.2.11.1.4	4.5.9.4
Noise cancelling	3.2.11.1.5	4.5.9.5
Supply line voltage and frequency variation	3.2	MIL-I-983
Shielding and radio frequency noise reduction	3.2	MIL-I-983
Temperature and humidity	3.2	MIL-I-983
Vibration	3.2	MIL-I-983
Shock	3.2	MIL-I-983

4.4 Acceptance inspection. - The equipment, selected as specified in 4.4.1.2, 4.4.1.3 and 4.4.1.4, shall be subjected to the following examination and tests:

Test	Requirement paragraph	Test paragraph
Group A		
General examination	3.2	MIL-I-983
Unit operating tests	3.2.12	4.7
Output voltage regulation	3.2.7.2	4.5.4
Frequency response	3.2.8.4.1	4.5.7.3
Distortion	3.2.8.4.1	4.5.7.4
Gain	3.2.8.4.1	4.5.7.5
Automatic gain control	3.2.8.4.1.3.5 (items(a)&(b) only)	4.5.7.6
Alarm signals	3.2.9.1	4.5.8
Mircophone impedance	3.2.11.1.1	4.5.9.1
Microphone frequency response	3.2.11.1.2	4.5.9.2
Microphone sensitivity	3.2.11.1.3	4.5.9.3
Group B		
System operation	3.2.12	4.6
Group C		
General examination	3.2	MIL-I-983
Cross talk	3.2.6.15	4.5.1
Switching transient noise	3.2.6.16	4.5.2
No load output voltage	3.2.7.1	4.5.3
Output voltage regulation	3.2.7.2	4.5.4

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Test	Requirement paragraph	Test paragraph
Group C		
Ripple voltage	3.2.7.3	4.5.5
Input power consumption	3.2.7.4	4.5.6
Amplifier test (conditions of)	3.2.8	4.5.7
Frequency response	3.2.8.4.1	4.5.7.3
Distortion	3.2.8.4.1	4.5.7.4
Power output	3.2.8.4.1	4.5.7.5
Automatic gain control	3.2.8.4.1.3.5	4.5.7.6
Rated load current	3.2.8.4.1	4.5.7.7
Cross talk (power line modulation)	3.2.8.4.1.4	4.5.7.8
Alarm signals	3.2.9.1	4.5.8
Microphone impedance	3.2.11.1.1	4.5.9.1
Microphone frequency response	3.2.11.1.2	4.5.9.2
Microphone sensitivity	3.2.11.1.3	4.5.9.3
Microphone endurance	3.2.11.1.4	4.5.9.4
Noise cancelling	3.2.11.1.5	4.5.9.5
System operation	3.2.12	4.6
Supply line voltage and frequency variation	3.2	MIL-I-983
Shielding and radio frequency noise reduction	3.2	MIL-I-983
Temperature and humidity	3.2	MIL-I-983
Vibration	3.2	MIL-I-983
Shock	3.2	MIL-I-983

4.4.1 Sampling for acceptance inspection. -

4.4.1.1 *Inspection lot.* - All equipment presented for delivery at one time shall be considered a lot. The lot may include the entire contract quanitity or it may be the production of any convenient time period.

4.4.1.2 Sampling for group A inspection. - All equipment shall be subjected to the group A examination and tests specified in 4.4, with acceptance based on 4.4.1.5.

4.4.1.3 Sampling for group B test. - Equipment components in the quantities specified (see 6.1) shall be subjected to the group B test specified in 4.4, with acceptance based on 4.4.1.5. The test shall be performed on equipment components presented for the first delivery of the contract. The test need be performed only once during the life of the production contract.

4.4.1.4 Sampling for group C examination and tests. - Group C examination and tests will be required when the basic design of the equipment or the material of a vital part has been changed. A production sample of each redesigned unit shall be selected and subjected to each applicable group C test specified in 4.4, with acceptance based on 4.4.1.6.

4.4.1.5 Acceptance - groups A and B inspection. - The results of each test shall be compared with this specification. Failure to conform to this specification for any test shall be counted as a defect and the equipment shall be considered a failure.

4.4.1.6 Acceptance - group C inspection. - The results of each test shall be compared with this specification. In the event of failure to conform to this specification for any group C test, the contractor shall correct the cause of faulure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.5 Test procedures. -

4.5.1 *Cross talk.* - Cross talk between audio circuits shall be measured to determine compliance with 3.2.6.15. Measurements shall be conducted with rated input at 1000 c.p.s. and rated output applied to the circuits under test. The ratio of the signal level on a transmitting circuit to the signal level recorded in adjacent circuits measured at the equipment terminals (expressed in decibels) will be the cross talk measurement. The circuits under measurement shall be connected to an equivalent of its normal load impedance.

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4.5.2 Switching transient noise. - Measurements shall be made during the operation of the equipment to determine compliance with 3.2.6.16.

4.5.3 No load output voltage. - This equipment shall be energized and the output voltage control rotated to determine conformance with 3.2.7.1.

4.5.4 Output voltage regulation. - Output voltage regulation shall be measured to determine conformance with 3.2.7.2.

4.5.5 Ripple voltage. - Ripple voltage shall be measured to determine conformance with 3.2.7.3.

4.5.6 Input power consumption. - Input power consumption shall be measured to determine conformance with 3.2.7.4.

4.5.7 Amplifier tests. - Amplifiers shall be tested at the following frequencies: 100, 200, 400, 1000, 2000, 4000 and 6000 cycles per second for frequency response, harmonic distortion and rated power output.

4.5.7.1 Test loads. - Tests shall be made on all amplifiers, with a noninductive resistance equal to the output impedance, connected to the output terminals.

4.5.7.2 Test voltages. - The voltage applied shall be the root mean square rated input voltage.

4.5.7.3 Frequency response. - With the specified load connected on the specified input (see 4.5.7.2) applied, the amplifier output shall be measured to determine compliance with 3.2.8.4.1.

4.5.7.4 Distortion. - With the specified load and input, the harmonic distortion shall be measured to determine compliance with 3.2.8.4.1.

4.5.7.5 Power output. - With the specified load and input, the power output shall be measured to determine compliance with 3.2.8.4.1.

4.5.7.6 Automatic gain control. - The characteristics of the automatic gain control circuits shall be measured to determine compliance with 3.2.8.4.1.3.

4.5.7.7 Rated load current. - Rated load current shall be measured to determine compliance with 3.2.8.4.1.

4.5.7.8 Cross talk due to power line modulation. - Tests shall be conducted to determine conformance with 3.2.8.4.1.4.

4.5.8 Alarm signals. - Alarm signals shall be recorded and submitted for the bureau or agency concerned approval in accordance with 3.2.9.1. A copy of the recorded alarm signals as approved by the bureau or agency concerned shall be supplied to the local inspection office. A playback reproducer shall be made available to the inspection office for playback of the approved recordings of the alarm signals. The local inspection office shall compare the alarm signals of production equipment against the approved alarms to insure that they are audibly intelligible and have the same characteristics.

4.5.9 Microphone M99/U tests. -

4.5.9.1 Microphone impedance. - Microphone impedance shall be measured to determine compliance with 3.2.11.1.1.

4.5.9.2 Frequency response. - A 150-ohm resistive load shall be connected across the output terminals for the microphone. The artificial mouth shall be adjusted to give a constant free field sound pressure of 100 microbars at a distance of 1/4 inch in front of and on the axis of the ring. The frequency shall be varied continuously from 200 to 6000 c.p.s. The microphone shall be set up facing the artificial mouth and so positioned that the mouthpiece will be coaxial with the ring of the artificial mouth and in a plane 1/4 inch in front of the ring. The voltage across the 150-ohm resistive load shall be measured to determine conformance with 3.2.11.1.2.

4.5.9.3 Microphone senvitivity. - Microphone sensitivity shall be measured to determine compliance with 3.2.11.1.3.

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4.5.9.4 Microphone endurance. - A variable frequency signal of 800 to 1250 c.p.s. at a voltage of 0.040 volts r.m.s. shall be impressed across the microphone voice coil circuit for an uninterrupted period of 100 hours of nominal room temperature of 75°F. The microphone shall then be tested to determine compliance with 3.2.11.1.4.

4.5.9.5 Noise cancelling. - The noise cancelling test shall be conducted in diffuse reverberation chamber:

(a) Thermal noise shall be applied to artificial voice and measured in 1/2 octave bands at 1/4 inch from lips of artificial head with a calibrated probe microphone.
(b) The artificial head shall be removed, but the probe microphone is left in position. Thermal noise shall be applied to loudspeakers in the diffuse reverberation chamber and 1/2 octave bands shall be measured from the noise cancelling microphone.
(c) The probe microphone shall be removed. The artificial head shall be replaced in its original position and the microphone to be tested shall be placed in the proper position.
(d) Thermal noise shall be applied to the artificial voice and 1/2 octave bands shall be measured from the noise cancelling microphone.
(e) Thermal noise shall be applied to the loudspeakers in the diffuse reverberation chamber and 1/2 octave bands shall be measured from the noise cancelling microphone.
(f) The values measured by the probe microphone shall be used to correct the values recorded at the output of the microphone being measured for actual conditions which exist at the time the measurements are made.
(g) Difference in corrected sensitivity is defined as the random noise discrimination for each 1/2 octave band. This shall be checked to determine conformance with 3.2.11.1.5.

4.6 System operations tests. - System operation test shall be conducted to determine that the equipment components shall function in accordance with the applicable portions of the overall system drawing (see 3.2.12.16). Only those quantities of components indicated by the asterisk (*) in the list of components (see 6.1(h)) shall be connected up to demonstrate the system capabilities. The following tests shall be included:

(a) Perform talk - Listen tests between intercom stations alone and then with headset-boom microphones.
(b) Perform talk - Listen tests on string circuits.
(c) Demonstrate operation of monitoring and muting facilities for loudspeakers.
(d) Demonstrate voice transmission priorities for each circuit.
(e) Demonstrate transmission of alarm signals, and priority of each alarm, and priority of alarms over voice transmission.
(f) Demonstrate test facilities and emergency reports (4 MC)

In demonstrating the communication capabilities of the system, the tests shall be conducted using power without battery, with battery only, and with battery and power supply. The power supply shall be loaded to draw 10 amperes.

4.6.1 Test equipment. - To demonstrate communications from the bridge station, a simulated portable ship's control unit shall be used. The unit shall consist of identical talk back speakers and wiring as shown in Drawing 815-1853054. The Sonar center operators' boxes shall require a switch to transfer operators' boxes to any of the positions on the break-in box. The Launch Preparation Panels stations shall require a switch to transfer between 35 MC groups 1 and 2. One talk-listen speaker (LS450-B) for test with the forward escape trunk station will be supplied by the Government. Also, one sound-powered telephone for test with emergency report circuit (4 MC) will be supplied.

4.7 Component operating tests. - Every component shall be tested on an individual basis for complete compliance with the applicable portions of 3.2.12.16. The systems test specified in 4.6 does not test all components submitted for shipment and does not necessarily test the entire capability of each type of equipment component.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts.-

5.1.1 Submarine intercommunication system, AN/WIC-(v)

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Shipboard repair parts.- Shipboard repair parts shall be preserved and packaged level A; packed level C and marked levels A and C, respectively, in accordance with Specification MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1).

5.2.1.1 Preservation, packaging, packing and marking.- The equipment, accessories, technical manuals, and stock repair parts shall be preserved and packaged level A or C; packed level A, B or C and marked levels A and B or C as specified (see 6.1) in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantity of items required.
(c) Quantity and type of manuals required (see Spec. MIL-M-15071).
(d) Supply demand control point for stock repair parts (see Spec. MIL-I-983).
(e) Whether unit stock number is to be entered on the identification plate (see MIL-I-983).
(f) Alarm signals and priority requirements.
(g) Requirement for nomenclature (nomenclature used in this specification will require subscript to identify manufacturer.)
(h) A list of stations, equipment components, and communication required at each station (see 4.6).
(i) Preservation and packaging, packing, and marking requirements if other than specified in 5.1 (see 5.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 23012, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

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Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 5830-N049Sh)

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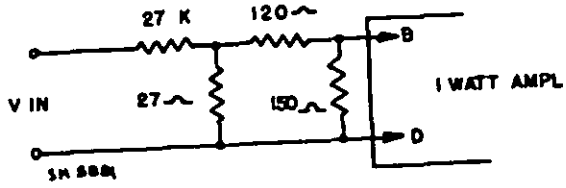


Figure 2 - Test connections and input pads for gain measurements of 1-Watt amplifiers.

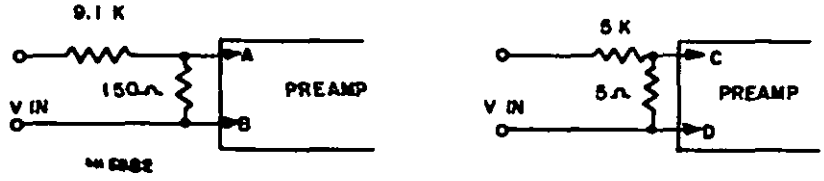


Figure 3 - Test connections and input pads for gain and frequency response measurements of the preamplifier.

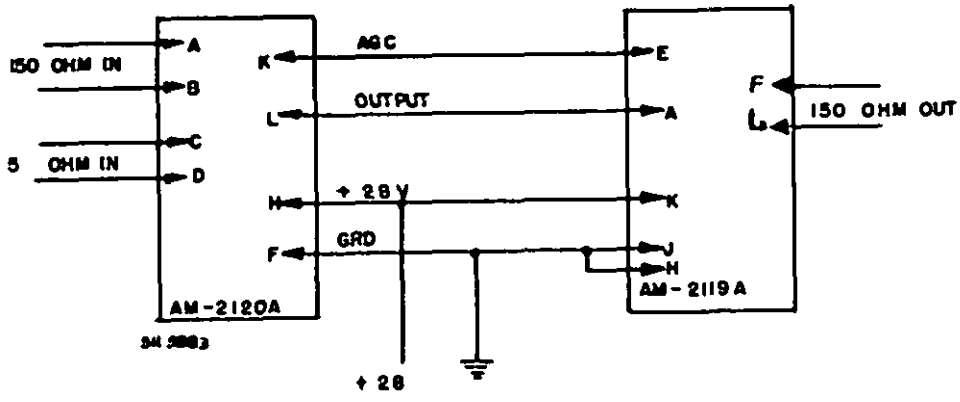


Figure 4 - Test connections for combined preamplifier and 1-Watt amplifier.

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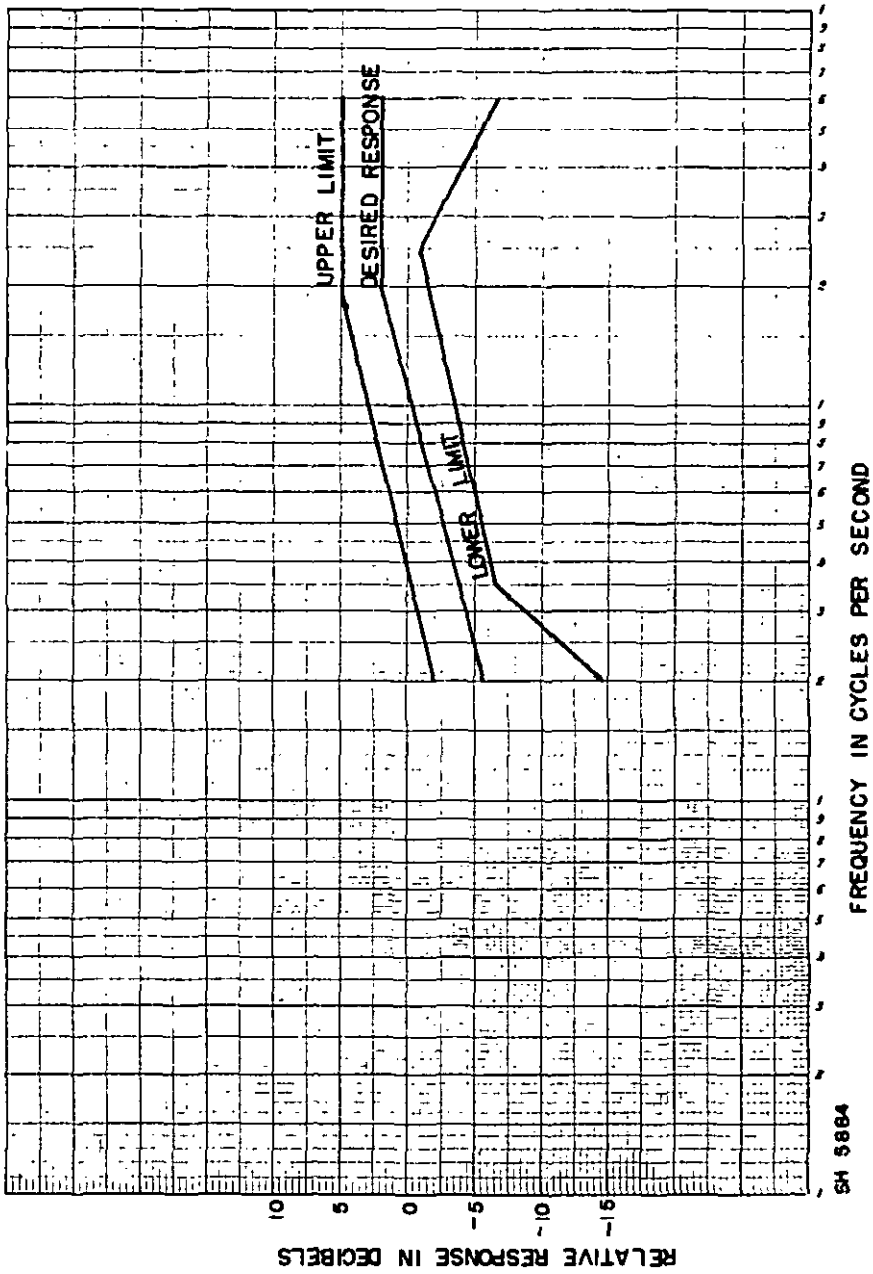


Figure 5 - Frequency response noise cancelling microphone M-99/U - limits.

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BUDGET BU. NO. 45-R309

SPECIFICATION ANALYSIS SHEET
NAVSHIPS-0883 (8-61)

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships.

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO
   IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification.) PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY (Print name and activity)	DATE

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MIL-S-23012A(SHIPS)
21 May 1962
SUPERSEDING
MIL-S-23012(SHIPS)
24 October 1961

MILITARY SPECIFICATION

SUBMARINE INTERCOMMUNICATION SYSTEM AN/WIC-(V)

NOTICE

Retain copies of Amendment-1 dated 25 October 1961 to MIL-S-23012(SHIPS).

Figure 1, sheets 1 through 10 issued under Amendment - 1 to Military Specification MIL-S-23012 dated 24 October 1961 forms a part of this specification as shown in paragraph 3.3.

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